

Weak Convergence of Point Processes

One of the fundamental results of probability theory is Poisson's limit theorem. It tells us that a sequence of binomial $\text{Bin}(p_n, n)$ distributed random variables B_n , $n = 1, 2, \dots$, converges in distribution to a Poisson random variable Y with parameter $\lambda > 0$ if and only if $np_n = EB_n \rightarrow EY = \lambda$. Here one deals with a rare event approximation because p_n , the success probability of the binomial random variable B_n , necessarily converges to zero. It is a remarkable result insofar that the distributional convergence relation $B_n \xrightarrow{d} Y$ is equivalent to the convergence of the expectations $EB_n \rightarrow EY$. We will see later that this property remains valid in a sense for the convergence in distribution of "binomial point processes" towards a Poisson random measure (PRM): convergence of the underlying mean measures of the "binomial point processes" implies their convergence in distribution towards a PRM with the limiting mean measure.

Any binomial $\text{Bin}(n, p)$ random variable B_n can be interpreted as the counting number of the successes in n independent trials with success probability $p \in (0, 1)$: if (X_i) is an iid sequence which describes an experiment with the success event A with probability p , then the number of successes

$$B_n = \sum_{i=1}^n I_A(X_i), \quad n \geq 1, \quad (9.0.1)$$

is binomially distributed with parameter (n, p) . This representation bears some resemblance to a point process. Indeed, we can introduce point processes

$$N_n = \sum_{i=1}^n \varepsilon_{X_i}, \quad n = 1, 2, \dots,$$

generated from the points $X_1, X_2, \dots$, on a suitable state space E . Then $B_n = N_n(A)$, and we can vary the event A over some σ -field $\mathcal{E}$. Since we know the notion of a PRM as a special point process it is reasonable to think

about distributional convergence of the “binomial processes” N_n to a PRM. Of course, since $EN_n(A) = nP(X_1 \in A) \rightarrow \infty$ or $= 0$ according as $P(X_1 \in A) > 0$ or $= 0$ we need to ensure that A depends on n and then $P(X_1 \in A_n) \rightarrow 0$ at a certain rate; we will often achieve this goal by considering the point process of the normalized and centered points $c_n^{-1}(X_i - d_n)$ for suitable constants $c_n > 0$ and $d_n \in \mathbb{R}$.

In what follows, we will deal with two major problems. *First*, point processes are random measures: each such random measure can be understood as a collection of random variables indexed by the sets of an appropriate σ -field which typically contains infinitely many elements. We have to clarify the meaning of convergence in distribution for these infinite-dimensional objects; see Section 9.1.1. *Second*, we need to explain the meaning of convergence of (possibly infinite) mean measures. Weak convergence of probability measures will be a guide. It will be one of the topics in Section 9.1.2, where we introduce the concept of *vague convergence* of measures. There we will also see that the convergence in distribution of point processes is equivalent to the pointwise convergence of the underlying Laplace functionals.

In Section 9.2 we apply the results of Section 9.1 in the context of extreme value theory. Extremes are important for applications in non-life insurance, in particular, in reinsurance. We study the point processes of exceedances and their weak convergence to a PRM. This convergence is equivalent to the convergence of maxima and upper order statistics for iid sequences. In Section 9.2.2 we characterize the possible limit distributions for maxima of iid random variables. The celebrated Fisher-Tippett Theorem (Theorem 9.2.7) summarizes the asymptotic theory for maxima. In Section 9.2.3 we consider maximum domains of attraction, i.e., we find conditions on a distribution F such that the normalized and centered partial maxima of an iid sequence with distribution F converge to a non-degenerate limit. In Section 9.2.4 we modify the point process of exceedances insofar that we assume that we consider a random sample indexed by a renewal process. The weak convergence of these processes is applied to reinsurance treaties of extreme value type as introduced in Section 3.4. In Section 9.3 we return to these treaties and study their limiting behavior.

9.1 Definition and Basic Examples

9.1.1 Convergence of the Finite-Dimensional Distributions

First of all we have to clarify:

What is the meaning of weak convergence of point processes?

This question cannot be answered at a completely elementary level. Consider point processes $N, N_1, N_2, \dots$ on the same state space $E \subset \mathbb{R}^d$. We know from Section 7.1.2 that the distribution of these point processes in $M_p(E)$, the space

of all point measures on E , is determined by their finite-dimensional distributions. Thus a natural requirement for weak convergence¹ of (N_n) towards N would be that, for any choice of “good” Borel sets $A_1, \dots, A_m \in \mathcal{E} = \mathcal{B}(E)$ and for any integer $m \geq 1$,

$$(N_n(A_1), \dots, N_n(A_m)) \xrightarrow{d} (N(A_1), \dots, N(A_m)) . \quad (9.1.2)$$

Every point process N_n can be considered as a stochastic process $(N_n(A))_{A \in \mathcal{E}}$ indexed by the sets $A \in \mathcal{E}$. From the theory of weak convergence of probability measures (see for example Billingsley [17]) we know that the convergence of the finite-dimensional distributions of stochastic processes is in general not sufficient for the weak convergence of their distributions: one needs an extra *tightness condition* meaning that the mass of the converging probability distributions should not disappear from compact sets.

Perhaps unexpectedly, point processes are user-friendly in the sense that tightness follows from the convergence of their finite-dimensional distributions; see for instance Daley and Vere-Jones [38], Theorem 9.1.VI. Hence we obtain quite an intuitive notion of weak convergence which is equivalent to the usual definition of weak convergence.

Definition 9.1.1 (Weak convergence of point processes)

Let $N, N_1, N_2, \dots$, be point processes on the state space $E \subset \mathbb{R}^d$ equipped with the σ -field $\mathcal{E}$ of its Borel sets. We say that the sequence of point processes (N_n) converges weakly to the point process N in $M_p(E)$ (we write $N_n \xrightarrow{d} N$) if (9.1.2) is satisfied for all possible choices of bounded sets $A_i \in \mathcal{E}$ satisfying $P(N(\partial A_i) = 0) = 1$, $i = 1, \dots, m$, $m \geq 1$, where ∂A denotes the boundary of A in E .

Assume for the moment that the state space E is an interval $(a, b] \subset \mathbb{R}$. Convergence of the finite-dimensional distributions can sometimes be checked by surprisingly simple means as the following result shows. Recall the notion of a *simple point process* from p. 218, i.e., the values of this process are simple point measures.

Theorem 9.1.2 (Kallenberg’s Theorem for weak convergence to a simple point process on an interval; see Kallenberg [79]; cf. Resnick [122], Proposition 3.22) *Let $N, N_1, N_2, \dots$, be point processes on $E = (a, b] \subset \mathbb{R}$ and assume that N is simple. Suppose the following two conditions hold:*

$$EN_n(c, d] \rightarrow EN(c, d], \quad a < c < d \leq b, \quad (9.1.3)$$

$$P(N_n(B) = 0) \rightarrow P(N(B) = 0), \quad (9.1.4)$$

¹ Instead of *weak convergence of point processes* it would be appropriate to refer to *weak convergence of the distributions of the point processes* or to *convergence in distribution of the point processes*; see Billingsley [17]. However, our usage is not uncommon in the literature; see for example Daley and Vere-Jones [38] and Resnick [122].

for any union $B = \cup_{i=1}^k (c_i, d_i]$ of disjoint intervals $(c_i, d_i] \subset (a, b]$ with $\min_{i=1, \dots, k} c_i > a$, $k \geq 1$. Then $N_n \xrightarrow{d} N$ in $M_p(E)$.

We apply Kallenberg's result to the weak convergence of Poisson processes.

Example 9.1.3 (Poisson processes converging to a simple Poisson process) We consider a sequence of Poisson processes N_n on $E = (a, b]$ with corresponding mean measures μ_n such that $N_n \xrightarrow{d} N$ in $M_p(E)$ for a simple Poisson process N with mean measure μ . Recall from Exercise 1 on p. 242 that a Poisson process N on $(a, b]$ is simple if and only if

$$\mu(\{x\}) = 0 \quad \text{for all } x \in (a, b]. \quad (9.1.5)$$

In particular, this condition is satisfied if μ has an intensity function $\lambda(x)$ on $(a, b]$.

Condition (9.1.3) of Kallenberg's Theorem then turns into

$$\mu_n(c, d] \rightarrow \mu(c, d] \quad \text{for } a < c < d \leq b. \quad (9.1.6)$$

Since the increments $N_n(c_i, d_i]$ for disjoint sets $(c_i, d_i]$ are independent condition (9.1.4) reads as

$$\begin{aligned} P(N_n(c, d] = 0) &= e^{-\mu_n(c, d]} \\ \rightarrow P(N(c, d] = 0) &= e^{-\mu(c, d]} \quad \text{for } a < c < d \leq b, \end{aligned} \quad (9.1.7)$$

which is equivalent to (9.1.6). Thus condition (9.1.6) is sufficient for $N_n \xrightarrow{d} N$.

Now assume that $N_n \xrightarrow{d} N$ in $M_p(E)$ for a simple PRM N . By Definition 9.1.1 we conclude that $N_n(c, d] \xrightarrow{d} N(c, d]$ for all $a < c < d \leq b$ such that $\mu(\partial(c, d]) = \mu(\{c, d\}) = 0$. The latter condition is satisfied in view of (9.1.5). Therefore (9.1.7) holds, implying (9.1.6).

We conclude that for PRMs $N, N_1, N_2, \dots$ on $(a, b]$ with mean measure μ of the limit process N satisfying (9.1.5), the relations $N_n \xrightarrow{d} N$ and (9.1.6) are equivalent. For homogeneous Poisson processes N_n and N with corresponding intensities λ_n and λ relation (9.1.6) turns into $\lambda_n \rightarrow \lambda$. $\square$

9.1.2 Convergence of Laplace Functionals

The Laplace functional (see p. 222) plays a similar role for point processes as characteristic functions for random variables. In particular, the continuity theorem of characteristic functions ensures that the distributional convergence of a sequence of random variables is equivalent to the pointwise convergence of the corresponding characteristic functions. The weak convergence of a sequence of point processes is equivalent to the convergence of their Laplace functionals indexed by a suitable family of bounded non-negative functions.

Such a set of functions is

$$\mathbb{C}_K^+(E) = \{g : g \text{ is a continuous, non-negative function on } E \\ \text{with compact support}\}.$$

Recall that a real-valued function g has compact support if there exists a compact set $K \subset E$ such that $g(x) = 0$ on K^c , the complement of K relative to E . We have already stressed on p. 223 that the distribution of a point process N on E is determined by its Laplace functional Ψ_N restricted to the functions $g \in \mathbb{C}_K^+(E)$. We can also test for weak convergence by checking Laplace functionals converge along $\mathbb{C}_K^+(E)$.

Theorem 9.1.4 (Criterion for weak convergence of point processes via convergence of Laplace functionals; see Daley and Vere-Jones [38], Proposition 9.1.VII, Resnick [122], Proposition 3.19.)

The sequence of point processes (N_n) converges weakly to the point process N in $M_p(E)$ if and only if the corresponding Laplace functionals converge for every $g \in \mathbb{C}_K^+(E)$:

$$\begin{aligned} \Psi_{N_n}(g) &= E \exp \left\{ - \int_E g dN_n \right\} \\ \rightarrow \Psi_N(g) &= E \exp \left\{ - \int_E g dN \right\}, \quad n \rightarrow \infty. \end{aligned} \quad (9.1.8)$$

Observe that $g \in \mathbb{C}_K^+(E)$ if and only if $zg \in \mathbb{C}_K^+(E)$ for every $z > 0$. Hence (9.1.8) for any $g \in \mathbb{C}_K^+(E)$ is equivalent to the convergence of the Laplace-Stieltjes transforms

$$E \exp \left\{ -z \int_E g dN_n \right\} \rightarrow E \exp \left\{ -z \int_E g dN \right\}, \quad z \geq 0.$$

In turn, this relation is equivalent to $\int_E g dN_n \xrightarrow{d} \int_E g dN$; see Feller [51], XIII 1, Theorem 2a.

Since we know the Laplace functional of a PRM it is straightforward to apply Theorem 9.1.4 to the weak convergence of PRMs towards a PRM. Also notice that we studied a special case of this convergence in Example 9.1.3, where we required the limiting process to be simple. This restriction is not needed below.

Example 9.1.5 (Convergence of PRMs towards a PRM)

From (7.2.11) recall the Laplace functional of a PRM(μ), denoted by N , on $E \subset \mathbb{R}^d$:

$$\Psi_N(g) = \exp \left\{ - \int_E \left(1 - e^{-g(x)} \right) \mu(dx) \right\}, \quad g \in \mathbb{C}_K^+(E).$$

We consider a sequence of PRMs N_n on E with corresponding Radon mean measures μ_n . According to Theorem 9.1.4, the relation $N_n \xrightarrow{d} N$ is equivalent to (9.1.8) for every $g \in \mathbb{C}_K^+(E)$:

$$\begin{aligned}\Psi_{N_n}(g) &= \exp \left\{ - \int_E (1 - e^{-g(x)}) \mu_n(dx) \right\} \\ \rightarrow \Psi_N(g) &= \exp \left\{ - \int_E (1 - e^{-g(x)}) \mu(dx) \right\} .\end{aligned}\quad (9.1.9)$$

In order to understand the latter relation consider a compact set $K \subset E$ with $\mu(\partial K) = 0$. Then a *formal* application of (9.1.9) to the indicator function $g = I_K$ yields

$$\begin{aligned}\mu_n(K) &= -(1 - e^{-1})^{-1} \log(\Psi_{N_n}(g)) \\ \rightarrow \mu(K) &= -(1 - e^{-1})^{-1} \log(\Psi_N(g)) .\end{aligned}$$

Unfortunately, the function I_K is not continuous, hence not in $\mathbb{C}_K^+(E)$. However, the condition $\mu(\partial K) = 0$ ensures that I_K is continuous with the exception of a set of μ -measure zero, namely the boundary of K . In what follows, we show that the convergence of the Laplace functionals (9.1.9) for $g \in \mathbb{C}_K^+(E)$ implies the relation

$$\mu_n(K) \rightarrow \mu(K) \quad \text{for any compact set } K \subset E \text{ satisfying } \mu(\partial K) = 0. \quad (9.1.10)$$

Before we continue with the proof we mention that the relation $N_n \xrightarrow{d} N$ immediately yields (9.1.10). Indeed, by the definition of $N_n \xrightarrow{d} N$ we have $N_n(A) \xrightarrow{d} N(A)$ for any bounded Borel set $A \subset E$ with the property

$$P(N(\partial A) = 0) = e^{-\mu(\partial A)} = 1, \quad \text{i.e.,} \quad \mu(\partial A) = 0,$$

and then in particular,

$$P(N_n(A) = 0) = e^{-\mu_n(A)} \rightarrow P(N(A) = 0) = e^{-\mu(A)} .$$

This implies (9.1.10).

A standard argument shows that one can approximate $g = I_K$ arbitrarily closely and uniformly by suitable functions $g_\delta \in \mathbb{C}_K^+(E)$ with support $K_\delta \downarrow K$ as $\delta \downarrow 0$ and such that $g \leq g_\delta \leq I_{K_\delta} \downarrow g$ and $g_\delta = g$ on K . Possible choices are

$$g_\delta(x) = 1 - \min(1, \rho(x, K)/\delta) \quad \text{and} \quad K_\delta = \{x \in E : \rho(x, K) \leq \delta\}, \quad (9.1.11)$$

where for any set A , $\rho(x, A) = \inf\{y \in A : |x - y|\}$. By construction of g_δ ,

$$g_\delta(x) = \begin{cases} 0 & \text{for } x \notin K_\delta, \\ g(x) & \text{for } x \in K. \end{cases}$$

Using this property, we obtain

$$\begin{aligned}
& |\log(\Psi_{N_n}(g)) - \log(\Psi_N(g))| \\
& \leq |\log(\Psi_{N_n}(g)) - \log(\Psi_{N_n}(g_\delta))| + |\log(\Psi_{N_n}(g_\delta)) - \log(\Psi_N(g_\delta))| \\
& \quad + |\log(\Psi_N(g_\delta)) - \log(\Psi_N(g))| \\
& = \int_{K_\delta \setminus K} \left(1 - e^{-g_\delta(x)}\right) \mu_n(dx) + |\log(\Psi_{N_n}(g_\delta)) - \log(\Psi_N(g_\delta))| \\
& \quad + \int_{K_\delta \setminus K} \left(1 - e^{-g_\delta(x)}\right) \mu(dx) \\
& \leq \int_{K_\delta \setminus K} \left(1 - e^{-g_\delta(x)}\right) \mu_n(dx) + |\log(\Psi_{N_n}(g_\delta)) - \log(\Psi_N(g_\delta))| + \mu(K_\delta \setminus K).
\end{aligned} \tag{9.1.12}$$

Continuity of the measure μ implies that $\mu(K_\delta \setminus K) \downarrow \mu(\emptyset) = 0$ as $\delta \downarrow 0$. Hence the third expression on the right-hand side can be made arbitrarily small. Theorem 9.1.4 yields $\Psi_{N_n}(g_\delta) \rightarrow \Psi_N(g_\delta)$ as $n \rightarrow \infty$ because $g_\delta \in \mathbb{C}_K^+(E)$. This makes the second expression vanish as $n \rightarrow \infty$. The first expression in (9.1.12) is bounded by $\mu_n(\overline{K_\delta \setminus K})$, where for any set $A \subset E$, $\overline{A}$ denotes its closure relative to E . For any $\epsilon > 0$ define $f_\epsilon(x) = 1 - \min(1, \rho(x, \overline{K_\delta \setminus K})/\epsilon)$. This function is in $\mathbb{C}_K^+(E)$ and has compact support $(\overline{K_\delta \setminus K})_\epsilon$. Similar arguments as above show that

$$\begin{aligned}
(1 - e^{-1}) \limsup_{n \rightarrow \infty} \mu_n(\overline{K_\delta \setminus K}) & \leq \lim_{n \rightarrow \infty} \int_E \left(1 - e^{-f_\epsilon(x)}\right) \mu_n(dx) \\
& = \int_E \left(1 - e^{-f_\epsilon(x)}\right) \mu(dx) \\
& \leq \mu((\overline{K_\delta \setminus K})_\epsilon).
\end{aligned}$$

Since $(\overline{K_\delta \setminus K})_\epsilon \downarrow \overline{K_\delta \setminus K}$ as $\epsilon \downarrow 0$ and $\overline{K_\delta \setminus K} \downarrow \partial K$ as $\delta \downarrow 0$ we conclude that

$$(1 - e^{-1}) \limsup_{n \rightarrow \infty} \mu_n(\overline{K_\delta \setminus K}) \leq \mu(\partial K) = 0.$$

Now collecting the arguments above, we conclude that (9.1.10) holds.

On the other hand, any $g \in \mathbb{C}_K^+(E)$ can be uniformly approximated by linear combinations of indicator functions I_K for suitable compact sets $K \subset E$ with $\mu(\partial K) = 0$. Using this uniform approximation and the relation (9.1.10) (which is equivalent to $\Psi_{N_n}(I_K) \rightarrow \Psi_N(I_K)$ for the sets K considered) one can show that (9.1.10) holds for any $g \in \mathbb{C}_K^+(E)$.

We conclude that for PRMs $N, N_1, N_2, \dots$ the weak convergence result $N_n \xrightarrow{d} N$ is equivalent to the convergence of the mean measures in the sense of (9.1.10). $\square$

In the previous example the weak convergence of PRMs towards a PRM is completely described by the convergence of their mean measures. We formalize this kind of convergence.

Definition 9.1.6 (Vague convergence of measures)

Let μ and μ_n , $n = 1, 2, \dots$, be Radon measures on $(E, \mathcal{E})$. We say that the sequence (μ_n) converges vaguely to μ if $\mu_n(A) \rightarrow \mu(A)$ for every relatively compact set $A \subset E$ such that $\mu(\partial A) = 0$. We write $\mu_n \xrightarrow{v} \mu$.

Vague convergence is similar to weak convergence of probability measures. Actually, weak and vague convergence coincide if μ_n and μ are probability measures. Another property which carries over from weak to vague convergence is the fact that it suffices to verify $\mu_n(A) \rightarrow \mu(A)$ for sets A which generate this kind of convergence. In particular, if the state space E is an interval of $\mathbb{R}$ it suffices to verify the condition $\mu_n(c, d] \rightarrow \mu(c, d]$ for all $(c, d] \subset E$ with $\mu(\{c, d\}) = 0$. Thus relation (9.1.6), which turned up in Example 9.1.3 as a necessary and sufficient condition for the weak convergence of Poisson processes towards a simple Poisson process, is equivalent to the vague convergence of the underlying mean measures.

Example 9.1.7 (Continuation of Example 9.1.5)

Assume that N_n is a PRM(μ_n) for a Radon measure μ_n on E for every $n = 1, 2, \dots$. It is an immediate consequence of the definition of the weak convergence relation $N_n \xrightarrow{d} N$ and the PRM property of the N_n 's that the random variables $N(A_1), \dots, N(A_m)$ are mutually independent and Poisson distributed for any choice of disjoint bounded sets $A_i \in \mathcal{E}$ satisfying $P(N(\partial A_i) = 0) = 1$, $i = 1, \dots, m$; see Exercise 1 on p. 299. Moreover, the corresponding Poisson parameters $\mu(A_i)$ of $N(A_i)$, $i = 1, \dots, m$, satisfy the additivity property:

$$\mu\left(\bigcup_{i=1}^m A_i\right) = \sum_{i=1}^m \mu(A_i).$$

Therefore a natural question arises:

Do the relation $N_n \xrightarrow{d} N$ and the PRM property of the point processes N_n imply that N is PRM(μ) for some mean measure μ ?

In other words, one needs to show that there exists a Radon measure μ on E that coincides with the additive measure μ on bounded sets A with the property $P(N(\partial A) = 0) = 1$. This fact is indeed true as a more advanced argument shows. Since $\mu_n(A) \rightarrow \mu(A)$ for all relatively compact sets $A \subset E$ with $\mu(\partial A) = 0$, Proposition 3.16 in Resnick [122] implies that the set of measures $\{\mu_n, n = 1, 2, \dots\}$ is vaguely relatively compact. This means that for any subsequence (n_k) of the positive integers there exists a subsequence (m_ℓ) (say) such that $\mu_{m_\ell} \xrightarrow{v} \tilde{\mu}$ for a Radon measure $\tilde{\mu}$ on E , possibly depending on the subsequence. However, for all such measures $\tilde{\mu}$, $\tilde{\mu}(K) = \mu(K)$ for any compact set K with $\mu(\partial K) = 0$ and therefore $\tilde{\mu} = \mu$. $\square$

Comments

The distribution of a point process is defined on the σ -field $\mathcal{M}_p(E)$ of subsets of $M_p(E)$. Therefore convergence in distribution of point processes is naturally defined as weak convergence of the distributions of the point processes on $\mathcal{M}_p(E)$. In order to avoid the infinite-dimensional setting and to keep the understanding of weak convergence at an intuitive level, we have chosen to define weak convergence via convergence of the finite-dimensional distributions. For our choice of state spaces, this definition is equivalent to the general notion of weak convergence of point processes; see Daley and Vere-Jones [38], Theorem 9.1.VI.

The weak convergence of point processes is treated in standard texts such as Kallenberg [79], Resnick [122, 124], Daley and Vere-Jones [38]. It plays a major role in the asymptotic theory of extremes for independent and strictly stationary sequences; see Leadbetter et al. [92], Resnick [122, 124], Embrechts et al. [46]. Moreover, the weak convergence of “binomial point processes” with iid points towards a PRM gives some justification for the use of general PRMs.

The importance of the concept of vague convergence for point measures was stressed in Kallenberg [79] and Resnick [121, 122] rather early on. It is one of the major tools for dealing with the weak convergence of point processes.

Exercises

Section 9.1.1

- (1) Let (N_n) be Poisson random measures on the state space $E \subset \mathbb{R}^d$ such that N_n has mean measure μ_n . Assume that $N_n \xrightarrow{d} N$ in the sense of Definition 9.1.1 and that N is a non-degenerate point process.
 - (a) Let $A \subset E$ be any bounded Borel set such that $P(N(\partial A) = 0) = 1$. Prove that there exists a number $\mu(A)$ such that $\mu_n(A) \rightarrow \mu(A)$ and that $\mu(A) = EN(A)$.
 - (b) Let $A_1, \dots, A_m \in \mathcal{E}$ be disjoint bounded Borel sets such that $P(N(\partial A_i) = 0) = 1$, $i = 1, \dots, m$. Show that the random vector $(N(A_1), \dots, N(A_m))$ has independent components and that $N(A_i)$ is $\text{Pois}(\mu(A_i))$ distributed, $i = 1, \dots, m$. Moreover, show that $\mu(\cup_{i=1}^m A_i) = \sum_{i=1}^m \mu(A_i)$.

Section 9.1.2

- (2) Consider Example 9.1.5.
 - (a) Show that for any compact set $K \subset E$, $I_K \leq g_\delta \leq I_{K_\delta} \downarrow I_K$ as $\delta \downarrow 0$ for g_δ as defined in (9.1.11).
 - (b) Show that any function $g \in \mathbb{C}_K^+(E)$ can be uniformly approximated by linear combinations of indicator functions I_K for suitable compact sets $K \subset E$ with $\mu(\partial K) = 0$.
- (3) Let (μ_n) be a sequence of probability measures on $(E, \mathcal{E})$.
 - (a) Let μ be a probability measure on $(E, \mathcal{E})$. Show that weak and vague convergence of (μ_n) towards μ are equivalent notions.

- (b) Give an example of a measure μ such that $\mu_n \xrightarrow{v} \mu$ but (μ_n) does not converge weakly.
- (4) Let X be a positive regularly varying random variable with distributional tail $\bar{F}(x) = x^{-\alpha}L(x)$, $x > 0$, where L is slowly varying and $\alpha > 0$. Choose any sequence $c_n \rightarrow \infty$ such that $n\bar{F}(c_n) \rightarrow 1$. Define the measures μ_n on $E = (0, \infty)$ by

$$\mu_n(x, \infty) = n\bar{F}(xc_n), \quad x > 0.$$

- Prove that $\mu_n \xrightarrow{v} \mu$ on E , where μ is given by $\mu(x, \infty) = x^{-\alpha}$, $x > 0$.
- (5) Let μ_n be the discrete uniform distribution on the set $\{n^{-1}, 2n^{-1}, \dots, 1\} \subset (0, 1]$. Show that $\mu_n \xrightarrow{v} \mu$ on E , where μ is the uniform $U(0, 1)$ distribution.

9.2 Point Processes of Exceedances and Extremes

9.2.1 Convergence of the Point Processes of Exceedances

In this section we consider one of the basic point processes which has major applications in extreme value theory: *the point process of exceedances*. We recall its definition from Example 7.1.6: for a sequence (X_i) of iid random variables with common distribution F and a sequence of thresholds $u_n \in \mathbb{R}$ the corresponding sequence of point processes of exceedances is given by

$$N_n = \sum_{i=1}^n \varepsilon_{n^{-1}i} I_{\{X_i > u_n\}}, \quad n = 1, 2, \dots \quad (9.2.13)$$

See Figure 9.2.2 for an illustration. Even though $E_n = \{n^{-1}, 2n^{-1}, \dots, 1\}$ is the support set of the measure N_n , it is more convenient, especially for asymptotics, to consider all N_n 's as measures on the same state space $E = (0, 1]$. As we learned in Example 7.1.6, the choice of these point processes is motivated by their relation with the maxima and order statistics of the samples $X_1, \dots, X_n$, $n = 1, 2, \dots$.

Our objective is to find conditions which ensure the weak convergence of the sequence of the point processes of exceedances (N_n) towards a homogeneous Poisson process N on $(0, 1]$. Of course, these conditions must be in terms of the threshold $u = u_n$ combined with the distribution F of the X_i 's.

Proposition 9.2.1 (Weak convergence of the point processes of exceedances) *Suppose that (X_n) is a sequence of iid random variables with common distribution F and, for a given sequence (u_n) of real numbers, (N_n) is the corresponding sequence of point processes given in (9.2.13). For any $\tau \in (0, \infty)$, the following statements are equivalent:*

- (1) *The limit $n\bar{F}(u_n) \rightarrow \tau$ exists.*
- (2) *The relation $N_n \xrightarrow{d} N$ holds in $M_p(E)$, where N is a homogeneous Poisson process on $E = (0, 1]$ with intensity τ .*

(3) For the sequence of partial maxima

$$M_n = \max_{i=1, \dots, n} X_i, \quad n = 1, 2, \dots,$$

the limit relation $P(M_n \leq u_n) \rightarrow e^{-\tau}$ holds.

(4) Consider the ordered sample²

$$X_{(1)} \leq \dots \leq X_{(n)} \quad (9.2.14)$$

of the iid sample $X_1, \dots, X_n$. For the sequence $(X_{(n-k+1)})$ of the k th largest order statistics and any fixed integer $k \geq 1$ the following limit relation holds:

$$\lim_{n \rightarrow \infty} P(X_{(n-k+1)} \leq u_n) = e^{-\tau} \sum_{i=0}^{k-1} \frac{\tau^i}{i!}. \quad (9.2.15)$$

Proof. (2) $\Leftrightarrow$ (1) First assume $N_n \xrightarrow{d} N$. In particular,

$$B_n = N_n(0, 1] \xrightarrow{d} N(0, 1] = Y.$$

This relation means that the sequence of the binomial $\text{Bin}(n, \bar{F}(u_n))$ random variables B_n converges in distribution to the $\text{Pois}(\tau)$ distributed random variable Y . By Poisson's limit theorem (or by a simple check of the converging characteristic functions or Laplace-Stieltjes transforms) the condition $n \bar{F}(u_n) \rightarrow \tau$ follows.

Now assume that $n \bar{F}(u_n) \rightarrow \tau$ holds. Since the limiting process N is assumed to be a homogeneous Poisson process, in particular simple, we may apply Kallenberg's Theorem 9.1.2.

For $(a, b] \subset (0, 1]$ the random variable

$$N_n(a, b] = \sum_{i=1}^n \varepsilon_{n^{-1}i}(a, b] I_{\{X_i > u_n\}} = \sum_{i: a < n^{-1}i \leq b} I_{\{X_i > u_n\}}$$

is $\text{Bin}(n(a, b], \bar{F}(u_n))$ distributed, where

$$n(a, b] = \#\{1 \leq i \leq n : na < i \leq nb\}.$$

Then, by the condition $n \bar{F}(u_n) \rightarrow \tau$,

$$EN_n(a, b] = n(a, b] \bar{F}(u_n) \sim (n(b-a)) \bar{F}(u_n) \sim (b-a)\tau = EN(a, b],$$

² The notation (9.2.14) for the ordered sample of $X_1, \dots, X_n$ is slightly imprecise. It would be more consequent to indicate the dependence of the order statistics on the sample size n , for example by using double indices. For the ease of presentation we avoid this approach, assuming implicitly that the order statistics have to be recalculated for every n .

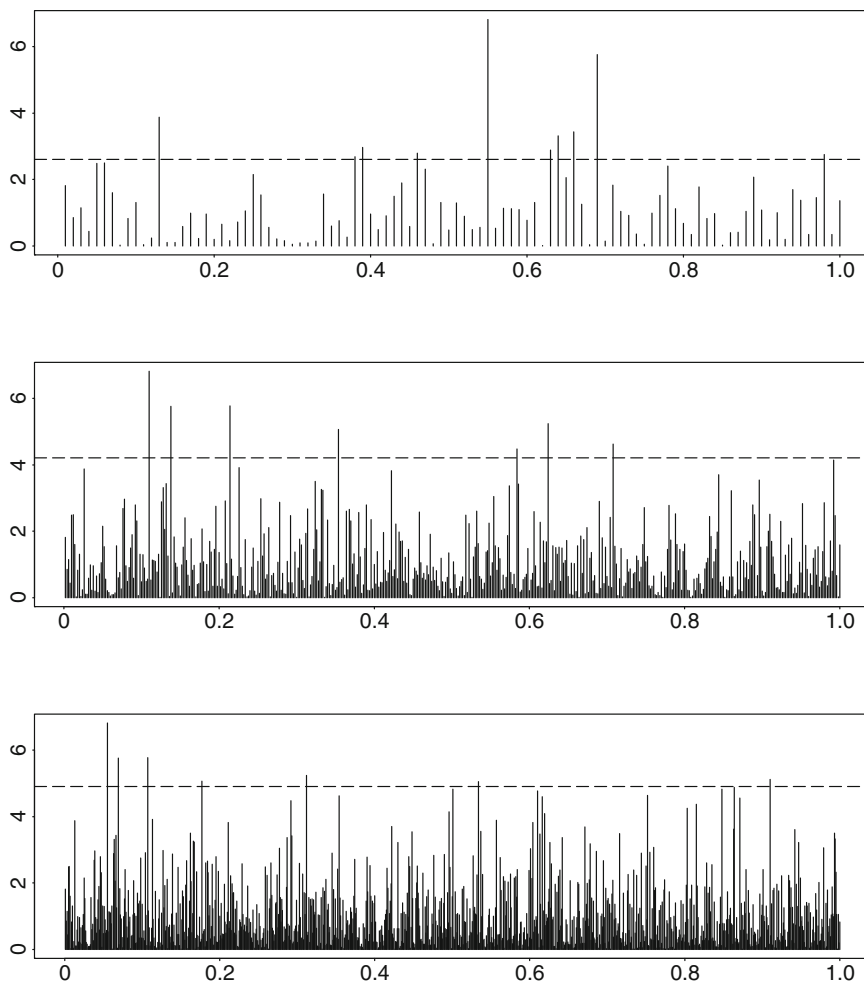


Figure 9.2.2 Visualization of the Poisson approximation for the exceedances of iid standard exponential random variables. The threshold increases with the sample size $n = 100$ (top), $n = 500$ (middle) and $n = 1000$ (bottom). Notice that the first sample also appears at the beginning of the second and the second at the beginning of the third.

which proves condition (9.1.3) of Kallenberg's Theorem. Here and in what follows, $f(x) \sim g(x)$ as $x \rightarrow x_0$ for positive f and g means that $f(x)/g(x) \rightarrow 1$ as $x \rightarrow x_0$.

Thus it remains to show the second condition (9.1.4). We start by observing that for any $0 < a < b \leq 1$,

$$\begin{aligned}
P(N_n(a, b] = 0) &= P\left(\max_{na < i \leq nb} X_i \leq u_n\right) \\
&= (F(u_n))^{n(a, b]} \sim (F(u_n))^{n(b-a)} \\
&= e^{(n(b-a)) \log(1-\bar{F}(u_n))} \\
&\rightarrow e^{-\tau(b-a)}.
\end{aligned} \tag{9.2.16}$$

In the last step we used the Taylor expansion $\log(1+x) = x(1+o(1))$ as $x \rightarrow 0$ and the fact that $n\bar{F}(u_n) \rightarrow \tau$ implies $\bar{F}(u_n) \rightarrow 0$.

Now recall the definition of the set B in (9.1.4). The independence of the random variables X_j and the disjointness of the sets $(c_i, d_i] \subset (0, 1]$, $c_i > 0$, which constitute the union B , imply that the counting variables $N_n(c_i, d_i]$ are independent. Therefore

$$\begin{aligned}
P(N_n(B) = 0) &= P(N_n(c_i, d_i] = 0, \quad i = 1, \dots, k) \\
&= \prod_{i=1}^k P(N_n(c_i, d_i] = 0) \\
&\rightarrow \prod_{i=1}^k e^{-\tau(d_i - c_i)}.
\end{aligned}$$

In the last step we made multiple use of (9.2.16). On the other hand, by the Poisson property of N ,

$$P(N(B) = 0) = e^{-\tau|B|} = e^{-\tau \sum_{i=1}^k (d_i - c_i)}.$$

This relation proves $N_n \xrightarrow{d} N$ by virtue of Kallenberg's Theorem 9.1.2.

(2) $\Rightarrow$ (3) Since $N_n \xrightarrow{d} N$ we have $N_n(0, 1] \xrightarrow{d} N(0, 1]$, and $N(0, 1]$ has a $\text{Pois}(\tau)$ distribution. Then it is immediate that

$$P(M_n \leq u_n) = P(N_n(0, 1] = 0) \rightarrow P(N(0, 1] = 0) = e^{-\tau}.$$

(3) $\Rightarrow$ (1) We have

$$P(M_n \leq u_n) = (F(u_n))^n = e^{-n \log(1-\bar{F}(u_n))} = e^{-n\bar{F}(u_n)(1+o(1))} \rightarrow e^{-\tau},$$

where we again used the properties $\bar{F}(u_n) \rightarrow 0$ and $\log(1+x) = x(1+o(1))$ as $x \rightarrow 0$. Now (1) is immediate.

In order to show the equivalence of (4) and one of the conditions (1)–(3) it suffices to consider the case that (4) holds for any fixed $k > 1$. Indeed, the case $k = 1$ coincides with (3).

(4) $\Rightarrow$ (1) First we observe that for any $k \geq 1$,

$$\{X_{(n-k+1)} \leq u_n\} = \{N_n(0, 1] < k\} \quad (9.2.17)$$

and that $N_n(0, 1]$ has a $\text{Bin}(n, \bar{F}(u_n))$ distribution. Therefore for any $k \geq 1$,

$$P(X_{(n-k+1)} \leq u_n) = \sum_{i=0}^{k-1} \binom{n}{i} [\bar{F}(u_n)]^i [F(u_n)]^{n-i}.$$

Since (9.2.15) is assumed for any fixed $k > 1$ the probability $F(u_n)$ necessarily converges to 1 as $n \rightarrow \infty$. Therefore as $n \rightarrow \infty$,

$$\begin{aligned} P(X_{(n-k+1)} \leq u_n) &\sim [F(u_n)]^n \sum_{i=0}^{k-1} \frac{[n \bar{F}(u_n)]^i}{i!} \\ &\sim e^{-n \bar{F}(u_n)} \sum_{i=0}^{k-1} \frac{[n \bar{F}(u_n)]^i}{i!}. \end{aligned} \quad (9.2.18)$$

Here we again made use of a Taylor expansion argument: $n \log(1 - \bar{F}(u_n)) \sim -n \bar{F}(u_n)$. The function

$$f_k(x) = e^{-x} \sum_{i=0}^{k-1} \frac{x^i}{i!}, \quad x > 0,$$

is continuous and decreasing, hence its inverse f_k^{-1} exists and is continuous. By virtue of (9.2.15) and (9.2.18) we have

$$f_k(n \bar{F}(u_n)) \rightarrow f_k(\tau).$$

By invertibility and continuity of f_k this implies that $n \bar{F}(u_n) \rightarrow \tau$, i.e., (1) holds.

(1) $\Rightarrow$ (4) The key relations are again (9.2.17) and (9.2.18). Since $n \bar{F}(u_n) \rightarrow \tau$ the limit relation (9.2.15) is immediate for any fixed $k \geq 1$.

This proves the proposition. $\square$

A short proof of the equivalence of (1) and (2) in Proposition 9.2.1 can be based on the equivalence of $N_n \xrightarrow{d} N$ and the convergence of the corresponding Laplace functionals $\Psi_{N_n}(f) \rightarrow \Psi_N(f)$. Indeed, for $f \in \mathbb{C}_K^+((0, 1])$ direct calculation shows that

$$\begin{aligned} \Psi_{N_n}(f) &= \prod_{i=1}^n \left(1 - \bar{F}(u_n) (1 - e^{-f(i/n)})\right) \\ &\rightarrow \Psi_N(f) = \exp \left\{ -\tau \int_{(0,1]} \left(1 - e^{-f(x)}\right) dx \right\}, \end{aligned} \quad (9.2.19)$$

if and only if $n \bar{F}(u_n) \rightarrow \tau$ for some $\tau \in (0, \infty)$. The reader is encouraged to verify this relation; see Exercise 2 on p. 322.

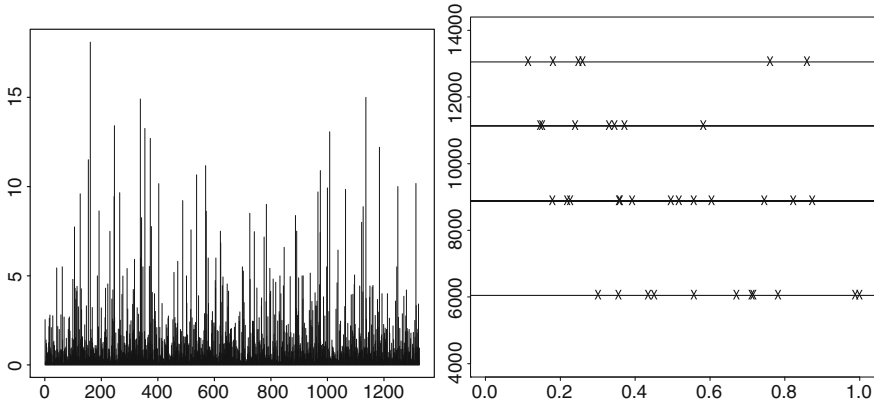


Figure 9.2.3 Left: *Insurance claim sizes (in 1000\$ units) caused by water damage, $n = 1762$ observations are available.* Right: *Visualization of the point processes of exceedances for this data set. For the sample size $[n/4]$ we choose the threshold $u_{[n/4]} = 6046$, correspondingly $u_{n/2} = 8880$, $u_{[3n/4]} = 11131$ and $u_n = 13051$. The exceedances times (renormalized to the interval $(0, 1]$) appear uniformly distributed.*

The Poisson approximation to the point processes of exceedances is visualized in Figure 9.2.3. We assume that these claim size data are iid. In view of the condition $n\bar{F}(u_n) \rightarrow \tau \in (0, \infty)$ the threshold u_n is raised in such a way that we roughly see the same number of exceedances for each threshold. According to Proposition 9.2.1 we may assume that the point process at the (renormalized) exceedance times i/n of the high threshold u_n is approximated by a homogeneous Poisson process. In agreement with the order statistics property of the homogeneous Poisson process we expect that the exceedance times occur uniformly on $(0, 1)$. This hypothesis is supported by the graph.

9.2.2 Convergence in Distribution of Maxima and Order Statistics Under Affine Transformations

In this section we study the links between the weak convergence of the point processes of the exceedances and the convergence in distribution of centered and normalized maxima and order statistics of iid samples when the sample size increases. Throughout we assume that (X_n) is a sequence of iid random variables with common distribution F . For a given sample $X_1, \dots, X_n$ consider the ordered sample

$$X_{(1)} \leq \dots \leq X_{(n)} = M_n.$$

First we investigate the convergence in distribution of the sequence of maxima (M_n) under affine transformations:

Which conditions on F ensure that there exist constants $c_n > 0$, $d_n \in \mathbb{R}$, $n = 1, 2, \dots$, and a non-constant limit random variable Y such that

$$c_n^{-1}(M_n - d_n) \xrightarrow{d} Y ? \quad (9.2.20)$$

If a non-degenerate limit distribution H in (9.2.20) exists it is called an *extreme value distribution*.

Example 9.2.4 (Three classes of extreme value distributions)

We consider the following three distributions:

Fréchet distribution: $\Phi_\alpha(x) = e^{-x^{-\alpha}}$ for some $\alpha > 0$, $x > 0$,

Weibull distribution: $\Psi_\alpha(x) = e^{-|x|^\alpha}$ for some $\alpha > 0$, $x < 0$,

Gumbel distribution: $\Lambda(x) = e^{-e^{-x}}$ for $x \in \mathbb{R}$.

Of course, $\Psi_\alpha(x) = 1$ for $x \geq 0$ and $\Phi_\alpha(x) = 0$ for $x \leq 0$ in order to ensure that Ψ_α and Φ_α are proper distributions. We also notice that Ψ_α is *not* the Weibull claim size distribution; see pp. 96 and 98. The latter distribution has support $(0, \infty)$. See Figure 9.2.5 for a visualization of the densities of these distributions.

Now assume that (X_i) is an iid sequence with common distribution H . Straightforward calculation yields for any $n \geq 1$,

$$n^{-1/\alpha} M_n \stackrel{d}{=} X_1 \quad \text{for } H = \Phi_\alpha,$$

$$n^{1/\alpha} M_n \stackrel{d}{=} X_1 \quad \text{for } H = \Psi_\alpha,$$

$$M_n - \log n \stackrel{d}{=} X_1 \quad \text{for } H = \Lambda.$$

Hence there exist constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that

$$c_n^{-1}(M_n - d_n) \stackrel{d}{=} X_1 \quad \text{for any } n \geq 1. \quad (9.2.21)$$

In particular, relation (9.2.20) holds trivially and therefore any of these distributions H is an extreme value distribution. Relation (9.2.21) remains valid for any distribution in the location-scale family $(H(ax + b))_{x \in \mathbb{R}}$, $a > 0$, $b \in \mathbb{R}$, for suitable choices of $c_n > 0$ and d_n ; see Exercise 6 on p. 322. We refer to this location-scale family as *the type of the distribution H* . Hence a distribution in the type of any of the distributions Φ_α , Ψ_α or Λ is an extreme value distribution as well. As a matter of fact, these types cover all extreme value distributions, as the celebrated Fisher-Tippett Theorem shows; see Theorem 9.2.7 below. Because of the validity of relation (9.2.21) for suitable constants $c_n > 0$ and $d_n \in \mathbb{R}$ for any $n \geq 1$, the extreme value distributions are also called *max-stable distributions*. $\square$

All extreme value distributions H are continuous. Therefore (9.2.20) can be written in the form

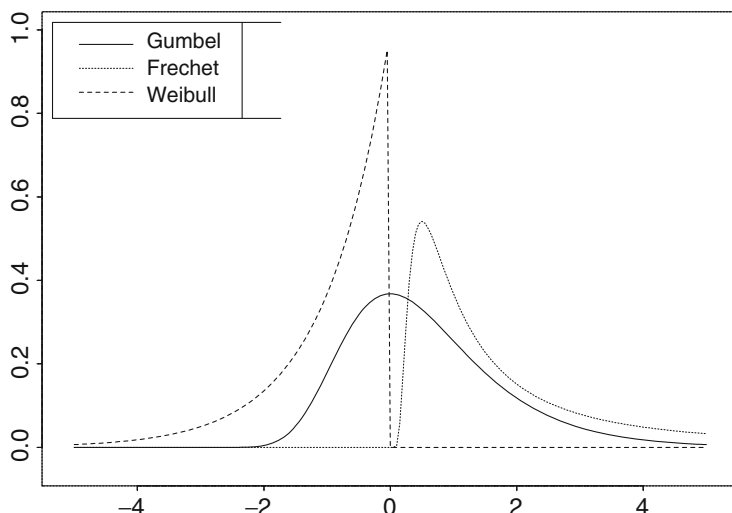


Figure 9.2.5 Densities of some extreme value distributions. We choose $\alpha = 1$ for the Fréchet and Weibull distributions.

$$\lim_{n \rightarrow \infty} P(M_n \leq c_n x + d_n) = H(x), \quad x \in \mathbb{R}.$$

It clearly suffices to consider x in the support of H such that $H(x) \in (0, 1)$. For these x 's, we can rewrite the latter relation as

$$\lim_{n \rightarrow \infty} P(M_n \leq u_n(x)) = e^{-\tau(x)},$$

where $u_n(x) = c_n x + d_n$ and $\tau(x) = -\log H(x)$. Now we are in the framework of Proposition 9.2.1.

Corollary 9.2.6 (Criterion for convergence in distribution of affinely transformed maxima and order statistics)

Let (X_i) be an iid sequence with common distribution F and H an extreme value distribution. Moreover, let $c_n > 0$ and $d_n \in \mathbb{R}$, $n \geq 1$, be given sequences of constants. Then the following statements are equivalent:

- (1) For any x in the support of H the following relation holds:

$$n \bar{F}(c_n x + d_n) \rightarrow -\log H(x).$$

- (2) For any x in the support of H the corresponding point processes of exceedances

$$N_{n,x} = \sum_{i=1}^n \varepsilon_{n^{-1}i} I_{\{X_i > c_n x + d_n\}}, \quad n = 1, 2, \dots,$$

converge in distribution to a homogeneous Poisson process on $(0, 1]$ with intensity $-\log H(x)$.

- (3) The sequence $(c_n^{-1}(M_n - d_n))$ converges in distribution to a random variable with distribution H .
- (4) For any x in the support of H and any fixed integer $k \geq 1$ the limit

$$\lim_{n \rightarrow \infty} P(c_n^{-1}(X_{(n-k+1)} - d_n) \leq x) = H(x) \sum_{i=0}^{k-1} \frac{(-\log H(x))^i}{i!}$$

exists and defines a proper distribution.

We have defined the extreme value distributions as the set of all non-degenerate limit distributions of centered and normalized maxima M_n of iid random variables X_i . In Example 9.2.4 we found that the Fréchet, Weibull and Gumbel distributions and their types are extreme value distributions. The following fundamental result shows that these are the only possible extreme value distributions.

Theorem 9.2.7 (Fisher-Tippett Theorem; cf. Embrechts et al. [46], Theorem 3.2.3, Resnick [122], Proposition 0.3)

Let (X_i) be a sequence of iid random variables. The relation $c_n^{-1}(M_n - d_n) \xrightarrow{d} Y$ holds for some constants $c_n > 0$, $d_n \in \mathbb{R}$ and a random variable Y with non-degenerate distribution H if and only if H is in the type of one of the distributions Φ_α , Ψ_α for some $\alpha > 0$, or Λ defined in Example 9.2.4.

The class of extreme value distributions can be written as a three-parameter family

$$H_{\xi; \mu, \sigma}(x) = \exp \left\{ - \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-1/\xi} \right\}. \quad (9.2.22)$$

Here $\xi \in \mathbb{R}$ is a shape parameter, $\mu \in \mathbb{R}$ and $\sigma > 0$ are location and scale parameters, respectively. The arguments x are chosen such that

$$1 + \xi \frac{x - \mu}{\sigma} > 0$$

in order to ensure that $H_{\xi; \mu, \sigma}$ defines a proper distribution. The case $\xi = 0$ is interpreted in the limiting sense as $\xi \rightarrow 0$ for fixed x :

$$H_{0; \mu, \sigma}(x) = \exp \left\{ -e^{-(x - \mu)/\sigma} \right\}, \quad x \in \mathbb{R}.$$

Obviously, the case $\xi = 0$ corresponds to the type of the Gumbel distribution Λ . The type of the Fréchet distribution Φ_α corresponds to $\xi = 1/\alpha > 0$ and the type of the Weibull distribution Ψ_α to $\xi = -1/\alpha < 0$. We leave it as Exercise 9 on p. 323 to verify that the types of Φ_α , Ψ_α and Λ are covered by the representation (9.2.22).

The joint three-parameter representation (9.2.22) of the three different types of the extreme value distributions is referred to as the *generalized extreme value distribution* which is often abbreviated as GEV. It is also referred to as the *Jenkinson-von Mises representation* of the extreme value distributions; see Section 3.4 in Embrechts et al. [46]. This representation was introduced for statistical applications. It allows one to estimate the parameters ξ , μ , σ by standard techniques such as the maximum likelihood method or the method of probability weighted moments in a unified way; see Embrechts et al. [46], Chapter 6.

9.2.3 Maximum Domains of Attraction

Consider a sequence (X_i) of iid random variables with common distribution F and assume that the relation

$$c_n^{-1}(M_n - d_n) \xrightarrow{d} Y \quad (9.2.23)$$

holds for suitable constants $c_n > 0$, $d_n \in \mathbb{R}$ and a random variable Y with an extreme value distribution H . Then the distribution F is said to be in the *maximum domain of attraction* of H ($F \in \text{MDA}(H)$). According to Corollary 9.2.6 the limit relation (9.2.23) is equivalent to the condition

$$n\bar{F}(c_n x + d_n) \rightarrow -\log H(x) \quad (9.2.24)$$

for x in the support of H . In various cases of interest it is possible to guess what the distribution H and the constants $c_n > 0$ and d_n are. However, a complete characterization of membership in $\text{MDA}(H)$ for a given extreme value distribution H is not trivial. We also mention that there exist distributions which do not belong to $\text{MDA}(H)$ for any extreme value distribution H . These include various well-known discrete distributions such as the Poisson, binomial, geometric, and negative binomial distributions; see Exercise 5(d) on p. 322.

It is clear from (9.2.24) that the constants c_n and d_n need to be expressed in terms of the right tail $\bar{F}$. This is not a very difficult task for F in the MDAs of the Fréchet and Weibull distributions, but it is rather sophisticated for the MDA of the Gumbel distribution. For complete descriptions of the MDAs, see for example Embrechts et al. [46], Chapter 3, or Resnick [122], Chapter 1. In these references one finds the following result and its proof. For its formulation recall the *quantile function* of the distribution F given by

$$F^{\leftarrow}(p) = \inf\{x \in \mathbb{R} : F(x) \geq p\}, \quad p \in (0, 1),$$

as well as the *right endpoint* of F :

$$x_F = \sup\{x \in \mathbb{R} : F(x) < 1\}.$$

We also recall that a measurable positive function L on $(0, \infty)$ is slowly varying if it satisfies $L(cx)/L(x) \rightarrow 1$ as $x \rightarrow \infty$ for any $c > 0$.

Theorem 9.2.8 (Characterization of MDAs)

Let (X_i) be an iid sequence with common distribution F .

- (1) The distribution $F \in \text{MDA}(\Phi_\alpha)$ for some $\alpha > 0$ if and only if

$$\bar{F}(x) = x^{-\alpha} L(x), \quad x > 0, \quad L \text{ slowly varying.}$$

The constants in (9.2.23) can be chosen such that

$$c_n = F^{\leftarrow}(1 - n^{-1}) \quad \text{and} \quad d_n = 0.$$

- (2) The distribution $F \in \text{MDA}(\Psi_\alpha)$ for some $\alpha > 0$ if and only if $x_F < \infty$ and

$$\bar{F}(x_F - x^{-1}) = x^{-\alpha} L(x), \quad x > 0, \quad L \text{ slowly varying.}$$

The constants in (9.2.23) can be chosen such that

$$c_n = x_F - F^{\leftarrow}(1 - n^{-1}) \quad \text{and} \quad d_n = x_F.$$

- (3) The distribution $F \in \text{MDA}(\Lambda)$ if and only if $F(\{x_F\}) = 0$ for $x_F < \infty$, and for $x_F \leq \infty$ there exist $x_0 < x_F$ and a positive absolutely continuous function $a(x)$, $x \in (x_0, x_F)$, such that the Lebesgue density $a'(x) \rightarrow 0$ as $x \uparrow x_F$ and

$$\bar{F}(x) \sim c \exp \left\{ - \int_{x_0}^x [a(t)]^{-1} dt \right\}, \quad x \uparrow x_F, \quad \text{for some constant } c > 0.$$

The constants in (9.2.23) can be chosen such that

$$d_n = F^{\leftarrow}(1 - n^{-1}) \quad \text{and} \quad c_n = a(d_n).$$

We refer to Tables 9.2.9, 9.2.10 and 9.2.14, where the MDA conditions and constants are summarized and the constants c_n, d_n are calculated for various distributions.

By its definition, a slowly varying function is positive for all $x > 0$. Therefore $F \in \text{MDA}(\Phi_\alpha)$ requires $x_F = \infty$. The MDA of the Fréchet distribution contains heavy-tailed distributions (those with a regularly varying tail), whereas the MDA of the Gumbel distribution ranges over a wide spectrum of distributional tails. It includes subexponential distributions (such as the log-normal distribution), but also light-tailed distributions with infinite right endpoint (such as the normal and the exponential distributions), and it also contains distributions with $x_F < \infty$. A distribution $F \in \text{MDA}(\Psi_\alpha)$ always satisfies $x_F < \infty$.

In what follows, we look at the MDAs in more detail and consider some prominent members in these classes of distributions.

The MDA of the Fréchet Distribution

This class of distributions contains various large claim distributions which we characterized as *heavy-tailed*; see for example Section 3.2.5. Recall that a (positive) regularly varying random variable X with tail index $\alpha > 0$ does not have finite moments of order greater than α . The MDA of Φ_α contains prominent members such as the Pareto, log-gamma, student, Cauchy, Burr and infinite variance stable distributions; see Table 9.2.9 for an overview.

The normalizing constants c_n in $\text{MDA}(\Phi_\alpha)$ can be derived in a rather simple way. Possible choices³ for c_n are the quantiles $F^{\leftarrow}(1 - n^{-1})$ or any sequence (c'_n) satisfying the asymptotic relation $n \bar{F}(c'_n) \rightarrow 1$. Of course, by virtue of the relation

$$n \bar{F}(c_n x + d_n) \rightarrow -\log \Phi_\alpha(x) = x^{-\alpha}, \quad x > 0, \quad (9.2.25)$$

setting $d_n = 0$ and $x = 1$, the quantities $c_n = F^{\leftarrow}(1 - n^{-1})$ satisfy the relation $n \bar{F}(c_n) \rightarrow 1$. We leave it as Exercise 11 on p. 323 to show that $c_n/c'_n \rightarrow 1$.

For a tail of the form $\bar{F}(x) = x^{-\alpha} L(x)$ for some $\alpha > 0$ and a slowly varying function L the MDA condition (9.2.25) is easily verified. Indeed, choose any sequence (c'_n) such that $n \bar{F}(c'_n) \rightarrow 1$. Then by regular variation of $\bar{F}(x)$ and since $c'_n \rightarrow \infty$,

$$n \bar{F}(c'_n x) \sim \frac{\bar{F}(c'_n x)}{\bar{F}(c'_n)} \sim x^{-\alpha} \frac{L(c'_n x)}{L(c'_n)} \rightarrow x^{-\alpha}, \quad x > 0.$$

If $\bar{F}(x) \sim K x^{-\alpha}$ for some positive constants $\alpha, K > 0$ it is not difficult to see that $(Kn)^{-1/\alpha} M_n \xrightarrow{d} Y$ where Y has distribution Φ_α . Such tail behavior is observed for the Pareto, Burr, infinite variance stable, Cauchy and student distributions. If the slowly varying function L in the tail $\bar{F}(x) = x^{-\alpha} L(x)$ is not asymptotically constant (as in the previously mentioned examples) the calculation of c_n is more complicated. However, it follows from the theory of regular variation (see for example Bingham et al. [19]) that $c_n = n^{1/\alpha} \ell(n)$ for a slowly varying function ℓ . This means that the shape of the sequence (c_n) is essentially determined by the powers $n^{1/\alpha}$. The slowly varying $\ell(n)$ is negligible in comparison to $n^{1/\alpha}$ since $\ell(n)/n^\delta \rightarrow 0$ and $\ell(n) n^\delta \rightarrow \infty$ for any $\delta > 0$; see Section 3.2.5.

³ In any limit relation $a_n^{-1}(A_n - b_n) \xrightarrow{d} Y$ the constants $a_n > 0$ and b_n are only determined up to asymptotic equivalence. Indeed, if $a'_n/a_n \rightarrow 1$ and $(b'_n - b_n)/a_n \rightarrow 0$ then we also have $(a'_n)^{-1}(A_n - b'_n) \xrightarrow{d} Y$. See also the Convergence to Types Theorem formulated on p. 323.

Fréchet distribution	$\Phi_\alpha(x) = e^{-x^{-\alpha}}, \quad x > 0, \quad \alpha > 0$
MDA (Φ_α)	$x_F = \infty, \quad \bar{F}(x) = x^{-\alpha} L(x), \quad L \text{ slowly varying}$
Constants	$c_n = F^{\leftarrow}(1 - n^{-1}) = n^{1/\alpha} \ell(n), \quad \ell \text{ slowly varying}, \quad d_n = 0$
Examples	
Cauchy	$f(x) = (\pi(1+x^2))^{-1}, \quad x \in \mathbb{R}$ $c_n = n/\pi$
Pareto, Burr, stable ($\alpha < 2$)	$\bar{F}(x) \sim K x^{-\alpha}, \quad x \rightarrow \infty, \quad K, \alpha > 0$ $c_n = (Kn)^{1/\alpha}$
Log-gamma	$f(x) = \frac{\alpha^\beta}{\Gamma(\beta)} (\log x)^{\beta-1} x^{-\alpha-1}, \quad x > 1, \quad \alpha, \beta > 0$ $c_n = [(\Gamma(\beta))^{-1} (\log n)^{\beta-1} n]^{1/\alpha}$

Table 9.2.9 Maximum domain of attraction of the Fréchet distribution. The symbol f stands for the Lebesgue density of the distribution.

The MDA of the Weibull Distribution

Distributions $F \in \text{MDA}(\Psi_\alpha)$ have finite right endpoint. Therefore they describe risks which potentially cannot get arbitrarily large. Such risks are bounded by a threshold such as claims covered by an excess-of-loss reinsurance treaty. There is also statistical evidence that the life span distribution of human beings is in the MDA of the Weibull distribution. In particular, extreme value methods yield rather convincing approximations to the empirical distribution of life span data at very old age; see Aarsen and de Haan [1] for a study of the life span of the Dutch population.

Prominent members of the MDA of the Weibull distribution are the uniform and beta distributions. Both exhibit power law behavior at the right endpoint of the distribution, i.e., $\bar{F}(x_F - x^{-1}) \sim K x^{-\alpha}$, and therefore the normalizing constants c_n can be determined from relation (9.2.24) which reads as

$$n \bar{F}(c_n x + x_F) \rightarrow -\log \Psi_\alpha(x) = |x|^\alpha, \quad x < 0.$$

Setting $x = -1$, we immediately see that we can choose $c_n = (Kn)^{-1/\alpha}$. If the slowly varying function in the tail $\bar{F}(x_F - x^{-1}) = x^{-\alpha} L(x)$ is not asymptotically constant, calculation of the constants c_n becomes more involved. See also Table 9.2.10 for an overview.

The MDAs of the Weibull and the Fréchet distributions are closely related because they are described by regular variation conditions on the tail of the

distribution at its right endpoint x_F . This analogy allows one to translate the limit theory in $\text{MDA}(\Phi_\alpha)$ to the corresponding limit theory in $\text{MDA}(\Psi_\alpha)$, and vice versa. For example, consider a sequence (X_i) of iid random variables with common distribution $F \in \text{MDA}(\Psi_\alpha)$ for some $\alpha > 0$. Then the transformed variables $Y_i = (x_F - X_i)^{-1}$ have tail $P(Y_1 > x) = L(x)x^{-\alpha}$, $x > 0$, for some slowly varying L , hence they are in the MDA of the Fréchet distribution Φ_α and then the limit theory in $\text{MDA}(\Phi_\alpha)$ can be applied to $\max_{i=1,\dots,n} Y_i$. The reader is encouraged to work out the details in Exercise 15 on p. 323.

Weibull distribution	$\Psi_\alpha(x) = e^{- x ^\alpha}$, $x < 0$, $\alpha > 0$
$\text{MDA}(\Psi_\alpha)$	$x_F < \infty$, $\bar{F}(x_F - x^{-1}) = x^{-\alpha} L(x)$, L slowly varying
Constants	$c_n = x_F - F^{\leftarrow}(1 - n^{-1}) = n^{-1/\alpha} \ell(n)$, ℓ slowly varying, $d_n = x_F$
Examples	
Uniform	$f(x) = 1$, $x \in (0, 1)$ $c_n = n^{-1}$, $d_n = 1$
Power law at x_F	$\bar{F}(x) \sim K(x_F - x)^\alpha$, $x \uparrow x_F$ $K, \alpha > 0$ $c_n = (Kn)^{-1/\alpha}$, $d_n = x_F$
Beta	$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}$, $0 < x < 1$, $a, b > 0$ $c_n = (n [\Gamma(a+b)/(\Gamma(a)\Gamma(b+1))])^{-1/b}$, $d_n = 1$

Table 9.2.10 Maximum domain of attraction of the Weibull distribution. The symbol f stands for the Lebesgue density of the distribution.

The MDA of the Gumbel Distribution

The MDA of the Gumbel distribution Λ contains a wide range of tail behaviors; see Table 9.2.14. It contains distributions with moderately heavy tails such as the log-normal distribution, the Weibull claim size distribution with tail $\bar{F}(x) = e^{-cx^\tau}$, $x > 0$, for constants $c > 0$, $\tau < 1$, and the Benktander distributions. These distributions are subexponential; see Section 3.2.6. In contrast to the claim size distributions in $\text{MDA}(\Phi_\alpha)$ the last mentioned distributions have finite moments of any order. However, their moment generating function does not exist as a consequence of their subexponentiality. But $\text{MDA}(\Lambda)$ also contains light-tailed distributions such as the gamma,

exponential, normal distributions and the Weibull claim size distribution with shape parameter $\tau \geq 1$ as well as certain distributions with finite right end-point.

Because of its wide range of tail behaviors the characterization of $\text{MDA}(\Lambda)$ is rather complicated. It contains all distributions F with $x_F \leq \infty$ whose tail has representation

$$\overline{F}(x) = c(x) \exp \left\{ - \int_{x_0}^x [a(t)]^{-1} dt \right\} \quad \text{for } x \in (x_0, x_F), \quad (9.2.26)$$

with $F(\{x_F\}) = 0$ for $x_F < \infty$. The functions $c(x)$ and $a(x)$ are positive such that $c(x) \rightarrow c$ for some constant $c \in (0, \infty)$, $a(x)$ has a Lebesgue density $a'(x)$ for $x \in (x_0, x_F)$ with the property that $a'(x) \rightarrow 0$ as $x \uparrow x_F$.

The function $a(x)$ is called an *auxiliary function*. It is not uniquely determined; there is a trade-off between the functions $c(x)$ and $a(x)$ possible. A possible choice of the auxiliary function is given by the *mean excess function*⁴

$$a(x) = E_F(X - x \mid X > x) = \frac{1}{\overline{F}(x)} \int_x^{x_F} \overline{F}(y) dy, \quad x < x_F.$$

An important subclass of $\text{MDA}(\Lambda)$ constitute the *von Mises functions*. They are distributions F where the function $c(x)$ in (9.2.26) can be chosen as a positive constant c . Most distributions in the MDA of the Gumbel distribution which are of interest in statistics or probability theory are von Mises functions. These include the Weibull claim size distribution, the normal, exponential, gamma distributions; see Table 9.2.12 for a list of von Mises functions and Table 9.2.14 for an overview of the MDA. Notice that von Mises distribution functions are smooth in the sense that they are absolutely continuous and monotone increasing on (x_0, x_F) . In contrast to von Mises functions a general distribution $F \in \text{MDA}(\Lambda)$ may have discontinuities or regions where it is constant. The following lemma shows that it actually suffices to focus the asymptotic extreme value theory on von Mises functions.

Lemma 9.2.11 (Tail equivalence of distributions in $\text{MDA}(\Lambda)$ and von Mises functions)

Assume that $F \in \text{MDA}(\Lambda)$ has representation (9.2.26) with $c(x) \rightarrow c > 0$ and G is the corresponding von Mises function with $c(x)$ replaced by c . Let (X_i) and (Y_i) be two sequences of iid random variables with common distributions F and G , respectively, and denote the corresponding partial maxima by $M_n(X)$ and $M_n(Y)$, $n = 1, 2, \dots$. Write Z for a Gumbel distributed random variable. Then there exist constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that $c_n^{-1}(M_n(X) - d_n) \xrightarrow{d} Z$ if and only if $c_n^{-1}(M_n(Y) - d_n) \xrightarrow{d} Z$.

Proof. Both distributions F and G have representation (9.2.26). Hence they are in $\text{MDA}(\Lambda)$. By Corollary 9.2.6, there exist constants $c_n > 0$ and d_n such that

⁴ See Section 3.2.3 for properties and interpretation of the mean excess function.

Distribution	Tail and auxiliary function
Exponential	$\bar{F}(x) = e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0$ $a(x) = \lambda^{-1}$
Weibull claim size distribution	$\bar{F}(x) = e^{-c x^\tau}, \quad x \geq 0, \quad c, \tau > 0$ $a(x) = c^{-1} \tau^{-1} x^{1-\tau}, \quad x > 0$
Erlang i.e., $\Gamma(n, \lambda)$	$\bar{F}(x) = e^{-\lambda x} \sum_{k=0}^{n-1} \frac{(\lambda x)^k}{k!}, \quad x \geq 0, \quad \lambda > 0, \quad n \geq 1$ $a(x) = \sum_{k=0}^{n-1} \frac{(n-1)!}{(n-k-1)!} \lambda^{-(k+1)} x^{-k}, \quad x > 0$
Exponential at finite x_F	$\bar{F}(x) = K e^{-\alpha/(x_F - x)}, \quad x < x_F < \infty, \quad \alpha, K > 0$ $a(x) = (x_F - x)^2 / \alpha, \quad x < x_F$
Benktander type I	$\bar{F}(x) = (1 + 2(\beta/\alpha) \log x) e^{-(\beta(\log x)^2 + (\alpha+1) \log x)},$ $x \geq 1, \quad \alpha, \beta > 0$ $a(x) = x/(\alpha + 2\beta \log x), \quad x \geq 1$
Benktander type II	$\bar{F}(x) = x^{-(1-\beta)} e^{-(\alpha/\beta)(x^\beta - 1)}, \quad x \geq 1, \quad \alpha > 0, 0 < \beta < 1$ $a(x) = \alpha^{-1} x^{1-\beta}, \quad x \geq 1$

Table 9.2.12 Von Mises functions F and their auxiliary functions $a(x)$.

$$n \bar{F}(c_n x + d_n) \rightarrow -\log \Lambda(x) = e^{-x}, \quad x \in \mathbb{R}. \quad (9.2.27)$$

In particular, $\bar{F}(c_n x + d_n) \rightarrow 0$ and therefore $c_n x + d_n \rightarrow x_F$. By assumption, $\bar{F}(y)/\bar{G}(y) \rightarrow 1$ as $y \uparrow x_F$ and therefore (9.2.27) holds if and only if

$$n \bar{G}(c_n x + d_n) \rightarrow -\log \Lambda(x) = e^{-x}, \quad x \in \mathbb{R}.$$

Equivalently, $c_n^{-1}(M_n(Y) - d_n) \xrightarrow{d} Z$. This concludes the proof. $\square$

For any distribution F with a positive density f we can write for suitable choices of $x_0 < x_F$ and constant $C > 0$,

$$\begin{aligned} \bar{F}(x) &= e^{\log \bar{F}(x)} \\ &= C e^{\int_{x_0}^x (\log \bar{F}(y))' dy} \\ &= C e^{-\int_{x_0}^x f(y)/\bar{F}(y) dy}, \quad x \in (x_0, x_F). \end{aligned}$$

By its definition, any von Mises function F has an a.e. positive Lebesgue density f on (x_0, x_F) , say. Then a possible choice of the auxiliary function $a(x)$ is given by⁵

$$a(x) = \bar{F}(x)/f(x), \quad x \in (x_0, x_F). \quad (9.2.28)$$

⁵ Note that $f(x)/\bar{F}(x)$ is the *hazard rate* of the distribution F ; see p. 108.

This equation follows from the a.e. uniqueness of the Radon-Nikodym derivative of a function (see Billingsley [18]). Moreover, if the function $a(x)$ defined in (9.2.28) has a Lebesgue density $a'(x)$ such that $a'(x) \rightarrow 0$ then necessarily $F \in \text{MDA}(\Lambda)$. Using representation (9.2.28), it is not difficult to identify the auxiliary functions of numerous von Mises functions.

The distributions $F \in \text{MDA}(\Lambda)$ have much lighter right tail than in the MDA of the Fréchet distribution. For $x_F < \infty$ this is straightforward. For $x_F = \infty$ the following property holds:

$$\lim_{x \rightarrow \infty} \frac{\overline{F}(xt)}{\overline{F}(x)} = \begin{cases} 0, & \text{if } t > 1, \\ \infty, & \text{if } 0 < t < 1. \end{cases} \quad (9.2.29)$$

It is called *rapid variation* of the tail $\overline{F}$ in contrast to regular variation.

We summarize some of the properties of von Mises functions:

Lemma 9.2.13 (Properties of von Mises functions)

Every von Mises function F is absolutely continuous on (x_0, x_F) , for some $x_0 < x_F$, with positive Lebesgue density f . The auxiliary function can be chosen as $a(x) = \overline{F}(x)/f(x)$. Moreover, the following properties hold.

(1) *If $x_F = \infty$ then $\overline{F}$ is rapidly varying in the sense of (9.2.29) and*

$$\lim_{x \rightarrow \infty} \frac{xf(x)}{\overline{F}(x)} = \infty.$$

(2) *If $x_F < \infty$ then (9.2.29) holds with $\overline{F}(x)$ replaced by $\overline{G}(x) = \overline{F}(x_F - x^{-1})$, $x > 0$, and*

$$\lim_{x \uparrow x_F} \frac{(x_F - x)f(x)}{\overline{F}(x)} = \infty.$$

We leave the proof as Exercise 17 on p. 324.

9.2.4 The Point Process of Exceedances at the Times of a Renewal Process

In a non-life insurance context one is interested in the magnitude of the largest claim sizes in a given period of time such as a year. The number of claims that arrive in this period will usually be considered as a random variable, and then we need to study the extremes of a random sample of claim sizes. The extreme value theory developed so far is not directly applicable to such objects. The purpose of this section is to introduce a modified point process of exceedances which takes into account random claim numbers and to study the corresponding random maxima and order statistics. The results will be applied to *reinsurance treaties of extreme value type* in Section 9.3. These treaties can be considered as functionals of the largest claim sizes in a random sample.

Gumbel	$\Lambda(x) = e^{-e^{-x}}, \quad x \in \mathbb{R}$
MDA(Λ)	$x_F \leq \infty, \quad \bar{F}(x) = c(x) \exp \left\{ - \int_{x_0}^x [a(t)]^{-1} dt \right\}, \quad x_0 < x < x_F,$ where $c(x) \rightarrow c > 0, \quad a(x) > 0, \quad a'(x) \rightarrow 0 \quad \text{as } x \uparrow x_F$
Constants	$d_n = F^{\leftarrow}(1 - n^{-1}), \quad c_n = a(d_n)$
Examples	
Exponential-like	$\bar{F}(x) \sim K e^{-\lambda x}, \quad x \rightarrow \infty, \quad K, \lambda > 0$ $c_n = \lambda^{-1}, \quad d_n = \lambda^{-1} \log(Kn)$
Weibull-like	$\bar{F}(x) \sim K x^\alpha e^{-c x^\tau}, \quad x \rightarrow \infty, \quad K, c, \tau > 0, \alpha \in \mathbb{R}$ $c_n = (c\tau)^{-1} (c^{-1} \log n)^{1/\tau-1},$ $d_n = (c^{-1} \log n)^{1/\tau} + c_n [\alpha \tau^{-1} \log(c^{-1} \log n) + \log K]$
Gamma	$f(x) = [\beta^\alpha / \Gamma(\alpha)] x^{\alpha-1} e^{-\beta x}, \quad x > 0, \quad \alpha, \beta > 0$ $c_n = \beta^{-1}, \quad d_n = \beta^{-1} [\log n + (\alpha - 1) \log \log n - \log \Gamma(\alpha)]$
Normal	$f(x) = (2\pi)^{-1/2} e^{-x^2/2}, \quad x \in \mathbb{R}$ $a_n = (2 \log n)^{1/2}, \quad c_n = a_n^{-1},$ $d_n = a_n - [\log(4\pi) + \log \log n] / (2 a_n)$
Log-normal	$f(x) = (2\pi)^{-1/2} (\sigma x)^{-1} e^{-(\log x - \mu)^2 / (2\sigma^2)}, \quad x > 0, \quad \mu \in \mathbb{R}, \sigma > 0$ $a_n = (2 \log n)^{1/2}, \quad c_n = \sigma a_n^{-1} d_n,$ $d_n = \exp \{ \mu + \sigma (a_n - [\log(4\pi) + \log \log n] / (2 a_n)) \}$
Exponential at $x_F < \infty$	$\bar{F}(x) = K e^{-\alpha/(x_F - x)}, \quad x < x_F, \quad \alpha, K > 0$ $c_n = \alpha [\log(Kn)]^{-2}, \quad d_n = x_F - \alpha [\log(Kn)]^{-1}$
Benktander type I	$\bar{F}(x) = (1 + 2(\beta/\alpha) \log x) e^{-(\beta(\log x)^2 + (\alpha+1) \log x)}, \quad x > 1, \alpha, \beta > 0$ $a_n = e^{-(\alpha+1)/(2\beta) + (\log n/\beta)^{1/2}}, \quad c_n = a_n / (2(\beta \log n)^{1/2}),$ $d_n = a_n \exp \left\{ [\log \log n + \log(4\beta/\alpha^2) + (\alpha+1)^2/(2\beta)] / (4(\beta \log n)^{1/2}) \right\}$
Benktander type II	$\bar{F}(x) = x^{-(1-\beta)} \exp \{ -(\alpha/\beta) (x^\beta - 1) \}, \quad x > 1, \quad \alpha > 0, 0 < \beta < 1$ $a_n = \beta \alpha^{-1} \log n, \quad c_n = \alpha^{-1} a_n^{1/\beta-1},$ $d_n = a_n^{1/\beta} + (\alpha\beta)^{-1} a_n^{1/\beta-1} \{ \alpha - (1-\beta) [\log \log n + \log(\beta/\alpha)] \}$

Table 9.2.14 Maximum domain of attraction of the Gumbel distribution. The symbol f stands for the Lebesgue density of the distribution.

The theory developed below parallels the results in the previous sections: we again exploit Kallenberg's Theorem. The advanced reader who feels comfortable with weak convergence of point processes may skip this section and immediately go to Section 9.3.

We describe the claim arrivals by the points of a renewal sequence

$$T_n = Y_1 + \cdots + Y_n, \quad n \geq 1,$$

with iid positive inter-arrival times $Y_1, Y_2, \dots$, and define the corresponding renewal process by

$$N_T(t) = \#\{i \geq 1 : T_i \leq t\}, \quad t \geq 0.$$

We mark every arrival T_n by a positive claim size X_n and assume that the sequences (T_n) and (X_n) are independent and (X_n) is iid.

A possible modification of the *point process of exceedances* for a given threshold u_n is then given by

$$\tilde{N}_n = \sum_{i=1}^{N_T(n)} \varepsilon_{n^{-1}T_i} I_{\{X_i > u_n\}}. \quad (9.2.30)$$

Here we again rescale time by n and count the exceedances of the increasing thresholds u_n by the random sample $X_1, \dots, X_{N_T(n)}$. As state space we choose the interval $E = (0, 1]$.

Notice that for any $0 < a < b \leq 1$,

$$\begin{aligned} \tilde{N}_n(a, b] &= \#\{i \geq 1 : na < T_i \leq nb, X_i > u_n\} \\ &= \#\{N_T(na) < i \leq N_T(nb) : X_i > u_n\}, \end{aligned}$$

i.e., $\tilde{N}_n(a, b]$ counts the number of exceedances of u_n by the claim sizes X_i that arrive in the interval $(na, nb]$.

The following result is analogous to Proposition 9.2.1 in the case of the standard point process of exceedances N_n .

Proposition 9.2.15 (Weak convergence of the point processes of exceedances at the times of a renewal process)

Suppose that (X_n) is a sequence of iid random variables with common distribution F and $\tilde{N}_n$ is the point process of the exceedances of the threshold u_n given in (9.2.30). We also assume that

$$\lambda^{-1} = EY_1 \in (0, \infty) \quad (9.2.31)$$

holds and that $n\overline{F}(u_n) \rightarrow \tau$ for some real number $\tau \in (0, \infty)$. Then the following statements are valid:

- (1) *The relation $\tilde{N}_n \xrightarrow{d} N$ holds in $M_p(E)$, where N is a homogeneous Poisson process on $E = (0, 1]$ with intensity $\lambda\tau$.*

(2) Consider the ordered sample

$$X_{(1)} \leq \cdots \leq X_{(N_T(n))}$$

of the random sample $X_1, \dots, X_{N_T(n)}$. For the sequence $(X_{(N_T(n)-k+1)})$ of the k th largest order statistics in a sample of size $N_T(n)$ and for any fixed integer $k \geq 1$ the following limit relation holds:

$$\lim_{n \rightarrow \infty} P(X_{(N_T(n)-k+1)} \leq u_n) = e^{-\lambda \tau} \sum_{i=0}^{k-1} \frac{(\lambda \tau)^i}{i!}.$$

Proof. We start by proving (1). Since the limit process is a homogeneous Poisson process it is simple. Therefore we may apply Kallenberg's Theorem 9.1.2. Then it suffices to show that

$$E\tilde{N}_n(a, b] \rightarrow EN(a, b] = \lambda \tau (b - a), \quad 0 < a < b \leq 1, \quad (9.2.32)$$

$$P(\tilde{N}_n(B) = 0) \rightarrow P(N(B) = 0), \quad (9.2.33)$$

for all sets $B = \cup_{i=1}^k (c_i, d_i]$ with disjoint intervals $(c_i, d_i] \subset (0, 1]$ such that $\min_{i=1, \dots, k} c_i > 0$, $k \geq 1$. Using the independence between N_T and (X_i) , we have

$$\begin{aligned} E\tilde{N}_n(a, b] &= E \left(\sum_{i=N_T(na)+1}^{N_T(nb)} I_{\{X_i > u_n\}} \right) \\ &= EN_T(na, nb] \bar{F}(u_n) \\ &= (n^{-1} EN_T(na, nb]) [n \bar{F}(u_n)] \\ &\rightarrow \lambda (b - a) \tau. \end{aligned}$$

In the last step we employed the assumption $n\bar{F}(u_n) \rightarrow \tau$ and the elementary renewal theorem $n^{-1} EN_T(nx) \rightarrow \lambda x$, $x > 0$; see Theorem 2.2.7. This proves (9.2.32).

Now we turn to the proof of (9.2.33). For simplicity we restrict ourselves to the set $B = (c_1, d_1] \cup (c_2, d_2]$, where $(c_i, d_i]$, $i = 1, 2$, are disjoint. The case of $k > 2$ disjoint sets $(c_i, d_i] \subset (0, 1]$ can be treated similarly. Conditioning on (T_i) and using the independence of (T_i) and (X_i) , we obtain

$$\begin{aligned} P(\tilde{N}_n(B) = 0) &= P \left(\max_{N_T(nc_1) < i \leq N_T(nd_1)} X_i \leq u_n, \max_{N_T(nc_2) < i \leq N_T(nd_2)} X_i \leq u_n \right) \\ &= E \left([F(u_n)]^{N_T(nc_1, nd_1) + N_T(nc_2, nd_2)} \right) \\ &= E \exp \{ n^{-1} (N_T(nc_1, nd_1) + N_T(nc_2, nd_2)) [n \log(1 - \bar{F}(u_n))] \} \end{aligned}$$

$$\begin{aligned} &\rightarrow \exp \left\{ -\lambda \tau \left((d_1 - c_1) + (d_2 - c_2) \right) \right\} \\ &= P(N(B) = 0). \end{aligned}$$

In this limit relation we applied the strong law of large numbers for renewal processes (Theorem 2.2.5) and relation $n\bar{F}(u_n) \rightarrow \tau$ in combination with Lebesgue dominated convergence. This proves (9.2.33) and, by Kallenberg's Theorem, the first statement.

Part (2) is an immediate consequence of (1). Indeed, for any $k \geq 1$,

$$\begin{aligned} P(X_{(N_T(n)-k+1)} \leq u_n) &= P(\tilde{N}_n(0, 1] < k) \\ &\rightarrow P(N(0, 1] < k). \end{aligned}$$

The random variable $N(0, 1]$ is $\text{Pois}(\lambda \tau)$ distributed which fact yields the form of the limit probability $P(N(0, 1] < k)$. This proves the proposition. $\square$

Proposition 9.2.15 enables one to derive the following analog of Corollary 9.2.6.

Corollary 9.2.16 *Let (X_i) be an iid sequence with common distribution F and $c_n > 0$, $d_n \in \mathbb{R}$, $n \geq 1$, be constants. For every $x \in \mathbb{R}$, consider the point process of the exceedances of the threshold $c_n x + d_n$:*

$$\tilde{N}_{n,x} = \sum_{i=1}^{N_T(n)} \varepsilon_{n^{-1}T_i} I_{\{X_i > c_n x + d_n\}}.$$

Assume that the relation $n\bar{F}(c_n x + d_n) \rightarrow -\log H(x)$ holds for any x in the support of an extreme value distribution H . Moreover, assume that (9.2.31) is satisfied. Then the following statements are valid:

- (1) *For any x in the support of H the sequence $(\tilde{N}_{n,x})$ of the point processes of exceedances converges in distribution to a homogeneous Poisson process on $(0, 1]$ with intensity $-\log([H(x)]^\lambda)$.*
- (2) *The sequence $(c_n^{-1}(M_{N_T(n)} - d_n))$ converges in distribution to a random variable with distribution H^λ .*
- (3) *For any x in the support of H and any integer $k \geq 1$ the limit*

$$\lim_{n \rightarrow \infty} P(c_n^{-1}(X_{(N_T(n)-k+1)} - d_n) \leq x) = [H(x)]^\lambda \sum_{i=0}^{k-1} \frac{(-\log([H(x)]^\lambda))^i}{i!}$$

exists and defines a proper distribution.

It is mentioned in Exercise 7 on p. 322 that for any extreme value distribution H , the power H^λ , $\lambda > 0$, is again an extreme value distribution. In particular, H and H^λ are of the same type as one of the standard extreme value distributions Φ_α , Ψ_α or Λ .

Comments

Weak convergence of point processes and random measures has been treated in all standard texts on the topic. We refer to Daley and Vere-Jones [38, 39, 40], Kallenberg [79], Matthes et al. [105], and Resnick [122, 123].

As mentioned above, a rigorous treatment of weak convergence of point processes requires one to consider them as random elements in an appropriate metric space. The general theory on weak convergence in metric spaces can be found for example in Billingsley [17]. One way to metrize weak convergence of point processes is via vague convergence of measures. Rigorous treatments are given in Kallenberg [79], Daley and Vere-Jones [38], Chapter 9, or Resnick [122], Chapter 3. Weak convergence of point processes and vague convergence are closely related to regular variation in $\mathbb{R}^d$; see Resnick [121, 122].

The weak convergence of the point processes of exceedances has been exploited in extreme value theory for a long time; see the texts Leadbetter et al. [92], Resnick [122], Falk et al. [48], Embrechts et al. [46]. In the case of an iid sequence the limiting point processes are homogeneous Poisson processes. The situation typically changes for the point processes of exceedances for strictly stationary sequences. Then the limiting processes often have multiple points located at the points of a homogeneous Poisson process. Such a process is also called a compound Poisson process. The fact that the limiting point process is not simple is caused by clusters of exceedances of high thresholds; due to the dependence in a stationary sequence exceedances of these thresholds occur not separated through time, but there is a tendency for large values to occur roughly at the same time. The extreme value theory for stationary sequences is still under development; the texts mentioned above treat the asymptotic theory for maxima and order statistics for various classes of strictly stationary sequences. Most of the recent theory is only available in journals.

Extreme value theory is a classical part of probability theory and statistics. In addition to the texts mentioned above, textbook treatments which address different levels of the theory and applications of extremes are the following ones: Gumbel [64], Galambos [52], Kotz and Nadarajah [88], Coles [34], Beirlant et al. [14, 13], de Haan and Ferreira [66], Reiss and Thomas [120], Resnick [124]. Treatments of multivariate extremes are given in Galambos [52], Resnick [121, 124], Kotz and Nadarajah [88], Coles [34], Falk et al. [48], de Haan and Ferreira [66].

Exercises

Section 9.2.1

- (1) Prove the following version of *Poisson's limit theorem*. Let (X_n) be an iid sequence of random variables with common distribution F , (u_n) be a sequence of real thresholds, $\tau \in (0, \infty)$. Then the relation $n\bar{F}(u_n) \rightarrow \tau$ holds if and only if the binomial $\text{Bin}(n, \bar{F}(u_n))$ random variable $B_n = \sum_{i=1}^n I_{\{X_i > u_n\}}$ converges in distribution to a $\text{Pois}(\tau)$ distributed random variable Y .

- (2) Prove by using Laplace functionals that the sequence of the point processes of exceedances N_n with thresholds u_n weakly converges to $\text{PRM}(\tau \text{ Leb})$, denoted by N , on $E = (0, 1]$ if and only if $n\bar{F}(u_n) \rightarrow \tau$ for some $\tau \in (0, \infty)$; see (9.2.19).
- (3) Assume that there exist constants u_n such that $n\bar{F}(u_n) \rightarrow \tau$ for some $\tau \in (0, \infty)$. Show that for any $\gamma \in (0, \infty)$ there exists a sequence $(u_n(\gamma))$ of constants such that $n\bar{F}(u_n(\gamma)) \rightarrow \gamma$.

Section 9.2.2

- (4) Let (X_i) be an iid sequence of random variables with common distribution F , (u_n) be a sequence of real numbers, $\tau \in [0, \infty]$ and $M_n = \max(X_1, \dots, X_n)$. Give a proof of the equivalence of the relations $P(M_n \leq u_n) \rightarrow e^{-\tau}$ and $n\bar{F}(u_n) \rightarrow \tau$ without point process arguments.
- (5) Let (X_i) be an iid sequence of random variables with common distribution F such that the right endpoint

$$x_F = \inf\{x : F(x) < 1\}$$

of the distribution F is finite. Assume that F has an atom at x_F , i.e., $\bar{F}(x_F) = F(x_F) - F(x_F-) > 0$.

- (a) Assume that $P(M_n \leq u_n) \rightarrow \rho$ for some $\rho \in [0, 1]$ and a sequence (u_n) . Show that necessarily $\rho = 0$ or $\rho = 1$.
- (b) Prove that there do not exist sequences of constants $c_n > 0$, $d_n \in \mathbb{R}$ and a non-degenerate limit distribution H such that $c_n^{-1}(M_n - d_n) \xrightarrow{d} Y$, where Y has distribution H .
- (c) The result of part (a) can be extended to general distributions F . Assume that $x_F \leq \infty$ and let $\tau \in (0, \infty)$. There exists a sequence (u_n) satisfying $n\bar{F}(u_n) \rightarrow \tau$ if and only if

$$\lim_{x \uparrow x_F} \frac{\bar{F}(x)}{\bar{F}(x-)} = 1, \quad (9.2.34)$$

and $F(x_F-) = 1$ in the case $x_F < \infty$. See Leadbetter et al. [92], Theorem 1.7.13, for a proof.

Now assume that $x_F = \infty$ and F has support on the integers. Show that condition (9.2.34) is equivalent to the relation

$$\lim_{n \rightarrow \infty} \frac{\bar{F}(n)}{\bar{F}(n-)} = 1.$$

- (d) Let (X_i) be an iid sequence with a common distribution F which is either binomial, Poisson, geometric or negative binomial. Show that there do not exist sequences of constants $c_n > 0$, $d_n \in \mathbb{R}$ and a non-degenerate distribution H such that $c_n^{-1}(M_n - d_n) \xrightarrow{d} Y$ for some random variable Y with distribution H .
- (6) Let H be any of the extreme value distributions Φ_α , Ψ_α or Λ . Show that the distribution $(H(ax + b))_{x \in \mathbb{R}}$ for any $a > 0$, $b \in \mathbb{R}$, is again an extreme value distribution; see Example 9.2.4.
- (7) Let H be an extreme value distribution and $\lambda > 0$. Show that H^λ is an extreme value distribution. In particular, show that H and H^λ are of the same type as one of the standard extreme value distributions Φ_α , Ψ_α or Λ .

- (8) Let X be a positive random variable with a Fréchet distribution Φ_α for some $\alpha > 0$. Show that $\log(X^\alpha)$ has distribution A , whereas $-X^{-1}$ has distribution Ψ_α .
- (9) Show that the Jenkinson-von Mises representation (9.2.22) covers the types of the extreme value distributions Φ_α , Ψ_α and A .
- (10) Let (X_i) be an iid sequence with common distribution F . Assume there exist constants $c_n > 0$ and d_n such that $c_n^{-1}(M_n - d_n) \xrightarrow{d} Y$ for some random variable Y with non-degenerate distribution H . Show that H is a *max-stable distribution*, i.e., for any $n \geq 1$ there exist $a_n > 0$ and b_n such that $a_n^{-1}(M_n(Y) - b_n) \stackrel{d}{=} Y$, where $Y, Y_1, Y_2, \dots$, constitute an iid sequence and $M_n(Y) = \max_{i=1, \dots, n} Y_i$. We refer to Theorem 3.2.2 in Embrechts et al. [46].

Hints: Show that the sequences $(c_{nk}^{-1}(M_{nk} - d_{nk}))$ and $(c_n^{-1}(M_{nk} - d_n))$ converge in distribution to some non-constant limits for any integer $k \geq 1$ and identify the limit distributions as a function of H . Then apply the *Convergence to Types Theorem* (see Bingham et al. [19], Lemma 8.0.1, Resnick [122], Embrechts et al. [46], Theorem A1.5):

Let $A, B, A_1, A_2, \dots$, be random variables and $r_n, r'_n > 0$, s_n, s'_n be constants. Assume that $r_n^{-1}(A_n - s_n) \xrightarrow{d} A$. Then the relation

$$(r'_n)^{-1}(A_n - s'_n) \xrightarrow{d} B \quad (9.2.35)$$

holds if and only if the limits $\lim_{n \rightarrow \infty} r_n/r'_n = r \in [0, \infty)$ and $\lim_{n \rightarrow \infty} (s_n - s'_n)/r'_n = s \in \mathbb{R}$ exist. If (9.2.35) holds then $B \stackrel{d}{=} rA + s$, and s, r are the unique constants for which this equality holds.

Section 9.2.3

- (11) Consider $F \in \text{MDA}(\Phi_\alpha)$ for some $\alpha > 0$. Show that any sequence (c'_n) satisfying the relation $n\overline{F}(c'_n) \rightarrow 1$ is asymptotically equivalent to $c_n = F^{\leftarrow}(1 - n^{-1})$, i.e., $c_n/c'_n \rightarrow 1$.
- (12) Let H be an extreme value distribution. Show that $H \in \text{MDA}(H)$.
- (13) The log-gamma distribution has density

$$f(x) = \frac{\alpha^\beta}{\Gamma(\beta)} (\log x)^{\beta-1} x^{-\alpha-1}, \quad x > 1, \quad \alpha, \beta > 0.$$

- (a) Determine the asymptotic behavior of $\overline{F}(x)$ as $x \rightarrow \infty$. Show that the log-gamma distribution is in the domain of attraction of the Fréchet distribution Φ_α . Hint: Use Karamata's theorem.
- (b) Determine the corresponding normalizing constants c_n .
- (14) The beta distribution has density

$$f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1} (1-x)^{b-1}, \quad 0 < x < 1, \quad a, b > 0.$$

- (a) Determine the asymptotic behavior of $\overline{F}(x)$ as $x \uparrow 1$. Show that the beta distribution is in the domain of attraction of the Weibull distribution Ψ_b .
- (b) Determine the corresponding constants c_n and d_n .
- (15) Let (X_i) be an iid sequence with common distribution $F \in \text{MDA}(\Psi_\alpha)$ for some positive α .

- (a) Show that the distribution of the iid random variables $Y_i = (x_F - X_i)^{-1}$ has right tail $\bar{F}_Y(x) = P(Y_1 > x) = x^{-\alpha} L(x)$, $x > 0$, for a slowly varying function L .
- (b) Use the fact that $F_Y \in \text{MDA}(\Phi_\alpha)$ to prove that $c_n^{-1}(M_n - x_F) \xrightarrow{d} Z$, where Z has distribution Ψ_α and $c_n = x_F - F^{\leftarrow}(1 - n^{-1})$.
- (16) Let $a(x)$ be an auxiliary function in the representation (9.2.26) of $F \in \text{MDA}(\Lambda)$. Show that $\lim_{x \rightarrow \infty} x^{-1}a(x) = 0$ if $x_F = \infty$ and $a(x) = o(x_F - x) = o(1)$ as $x \uparrow x_F$ if $x_F < \infty$.
- (17) Prove Lemma 9.2.13.
- (18) Let (X_i) be a sequence of iid random variables with common distribution F which is a von Mises function. Show that there exist constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that $c_n^{-1}(M_n - d_n) \xrightarrow{d} Y$, where Y has distribution Λ . The constants are given by $c_n = F^{-1}(1 - n^{-1})$ and $c_n = a(d_n)$, where F^{-1} is the inverse of F and $a(x)$ is the auxiliary function in representation (9.2.26).

Hints: First show that

$$\lim_{x \uparrow x_F} \frac{\bar{F}(x + a(x)t)}{\bar{F}(x)} = e^{-t}, \quad t \in \mathbb{R},$$

by using the form of the von Mises functions given in (9.2.26). Then choose c_n such that $\bar{F}(c_n) = n^{-1}$ and apply Corollary 9.2.6.

Section 9.2.4

- (19) Consider an iid sequence (Y_i) of positive random variables with finite mean $\lambda^{-1} = EY_1$ and the corresponding renewal sequence $T_n = Y_1 + \cdots + Y_n$, $n \geq 1$. Prove the uniform strong law of large numbers

$$\sup_{x \in [0, 1]} \left| \frac{T_{[nx]}}{n} - \frac{x}{\lambda} \right| \xrightarrow{\text{a.s.}} 0.$$

9.3 Asymptotic Theory for the Reinsurance Treaties of Extreme Value Type

In this section we consider the same setting as in Section 9.2.4, i.e., we consider an iid claim size sequence (X_n) of positive random variables with common distribution F and a renewal claim arrival sequence (T_n) which is independent of (X_n) . We again write

$$N_T(t) = \#\{i \geq 1 : T_i \leq t\}, \quad t \geq 0,$$

for the corresponding renewal counting process. For ease of presentation, we restrict ourselves to a homogeneous Poisson process N_T with intensity λ . However, the asymptotic theory developed below can be extended to the case of general renewal processes N_T with $\lambda^{-1} = EY_1 \in (0, \infty)$; see Embrechts et al. [46], Section 8.7.

Reinsurance treaties of extreme value type (see Section 3.4), in contrast to *treaties of random walk type*, only involve the largest claims in the random sample $X_1, \dots, X_{N_T(t)}$ which have occurred up to time t .

- (1) *Largest claims reinsurance*. In this treaty the reinsurer covers the claim amount

$$R_{LC}(t) = \sum_{i=1}^k X_{(N_T(t)-i+1)}$$

in the period $[0, t]$, $t \geq 0$, for a fixed integer $k \geq 1$.

- (2) *ECOMOR reinsurance* (*Excédent du coût moyen relatif*). This treaty has the form of an excess-of-loss reinsurance treaty with a random deductible which is determined by the k th largest claim in the portfolio. The reinsurer covers the claim amount

$$\begin{aligned} R_{\text{ECOMOR}}(t) &= \sum_{i=1}^{N_T(t)} (X_{(N_T(t)-i+1)} - X_{(N_T(t)-k+1)})_+ \\ &= \sum_{i=1}^{k-1} X_{(N_T(t)-i+1)} - (k-1)X_{(N_T(t)-k+1)} \end{aligned} \quad (9.3.36)$$

in the period $[0, t]$, $t \geq 0$, for a fixed integer $k \geq 2$.

Example 9.3.1 (The joint distribution of the k largest claim sizes)

The distribution of $X_{(N_T(t)-k+1)}$ is easily calculated if N_T is a homogeneous Poisson process with intensity $\lambda > 0$. Indeed, writing N for the marked PRM of the points (T_i, X_i) with mean measure $\lambda \text{Leb} \times F$ and state space $E = (0, \infty)^2$, the Poisson property immediately yields for the k th largest claim size in the period $(0, t]$,

$$\begin{aligned} P(X_{(N_T(t)-k+1)} \leq x) &= P(N((0, t] \times (x, \infty)) < k) \\ &= \sum_{i=0}^{k-1} e^{-\lambda t \bar{F}(x)} \frac{(\lambda t \bar{F}(x))^i}{i!}. \end{aligned}$$

We have already derived this formula in Example 3.4.2 by some elementary considerations. At this point we can use advanced point process theory and gain much more information about the joint distribution of the largest claim sizes. For example, by a geometric argument it is also possible to determine the joint distribution of the vector $(X_{(N_T(t)-k+1)}, \dots, X_{(N_T(t))})$. Indeed, for $x_k < \dots < x_1$, $k \geq 1$, the sets

$$\begin{aligned} A_k &= (0, t] \times (x_k, x_{k-1}], \dots, A_2 = (0, t] \times (x_2, x_1], \\ A_1 &= (0, t] \times (x_1, \infty), \end{aligned}$$

are disjoint and therefore the PRM property implies that the counting variables $K_i = N(A_i)$, $i = 1, \dots, k$, are mutually independent. Hence for any integers $\ell_i \geq 0$, $i = 1, \dots, k$,

$$P(K_1 = \ell_1, \dots, K_k = \ell_k) = P(K_1 = \ell_1) \cdots P(K_k = \ell_k),$$

and then

$$\begin{aligned}
& P(X_{(N_T(t)-k+1)} \leq x_k, \dots, X_{(N_T(t))} \leq x_1) \\
&= P(N((0, t] \times (x_k, \infty)) < k, \dots, N((0, t] \times (x_1, \infty)) < 1) \\
&= P(K_k + \dots + K_1 < k, K_{k-1} + \dots + K_1 < k-1, \dots, K_1 < 1) \\
&= P(K_k + \dots + K_2 \leq k-1, \dots, K_3 + K_2 \leq 2, K_2 \leq 1, K_1 = 0).
\end{aligned}$$

The latter relation can be evaluated by some elementary calculations. For example, for $x_2 < x_1$,

$$\begin{aligned}
& P(X_{(N_T(t)-1)} \leq x_2, X_{(N_T(t))} \leq x_1) \\
&= P(K_1 = 0, K_2 \leq 1) \\
&= P(K_1 = 0) [P(K_2 = 0) + P(K_2 = 1)] \\
&= e^{-\lambda t \bar{F}(x_1)} \left[e^{-\lambda t F(x_2, x_1)} (1 + \lambda t F(x_2, x_1)) \right] \\
&= e^{-\lambda t \bar{F}(x_2)} (1 + \lambda t F(x_2, x_1)),
\end{aligned}$$

and for $x_3 < x_2 < x_1$,

$$\begin{aligned}
& P(X_{(N_T(t)-2)} \leq x_3, X_{(N_T(t)-1)} \leq x_2, X_{(N_T(t))} \leq x_1) \\
&= P(K_1 = 0, K_2 \leq 1, K_2 + K_3 \leq 2) \\
&= P(K_1 = 0) \sum_{\ell_2=0}^1 P(K_2 = \ell_2) \sum_{\ell_3=0}^{2-\ell_2} P(K_3 = \ell_3) \\
&= e^{-\lambda t \bar{F}(x_3)} \left(1 + \lambda t F(x_3, x_1) + (\lambda t)^2 F(x_3, x_2) \left(\frac{1}{2} F(x_3, x_1) + F(x_2, x_1) \right) \right).
\end{aligned}$$

Obviously, these calculations become even more tedious the larger k :

$$\begin{aligned}
& P(X_{(N_T(t)-k+1)} \leq x_k, \dots, X_{(N_T(t))} \leq x_1) \\
&= P(K_1 = 0, K_2 \leq 1, K_2 + K_3 \leq 2, \dots, K_2 + \dots + K_k \leq k-1) \\
&= \sum_{\ell_2=0}^1 \sum_{\ell_3=0}^{2-\ell_2} \sum_{\ell_4=0}^{3-\ell_3-\ell_2} \dots \sum_{\ell_k=0}^{k-1-\ell_{k-1}-\dots-\ell_2} \prod_{l=0}^k P(K_l = \ell_l) \\
&= e^{-\lambda t \bar{F}(x_k)} \sum_{\ell_2=0}^1 \sum_{\ell_3=0}^{2-\ell_2} \sum_{\ell_4=0}^{3-\ell_3-\ell_2} \dots \sum_{\ell_k=0}^{k-1-\ell_{k-1}-\dots-\ell_2} \prod_{l=0}^k \frac{(\lambda t F(x_l, x_{l-1}))^{\ell_l}}{\ell_l!}.
\end{aligned} \tag{9.3.37}$$

□

Because we explicitly know the joint distribution of the vector

$$(X_{(N_T(t)-k+1)}, \dots, X_{(N_T(t))})$$

of the largest claim sizes in a portfolio we can *in principle* calculate the distribution of any measurable function of this vector, in particular, the distributions of $R_{LC}(t)$ and $R_{ECOMOR}(t)$. However, despite the “simplicity” of the structure of the functions $R_{LC}(t)$ and $R_{ECOMOR}(t)$, they are still too complicated in order to be tractable for general distributions F . In order to better understand the distribution of these reinsurance treaties asymptotic extreme value techniques as considered in the previous sections can be employed. We illustrate this aspect in the remainder of this section.

In what follows, we assume that the claim size distribution F belongs to the maximum domain of attraction (MDA) of some extreme value distribution H . We conclude from Corollary 9.2.16 that for each of the order statistics constituting the quantities $R_{LC}(n)$ and $R_{ECOMOR}(n)$ we have an asymptotic extreme value theory at our disposal. However, in order to evaluate the asymptotic behavior of these treaties we need to know the joint limit distribution of the vectors $c_n^{-1}(X_{(N_T(n)-k+1)} - d_n, \dots, X_{(N_T(n))} - d_n)$ for suitable normalizing and centering constants $c_n > 0$ and $d_n \in \mathbb{R}$, $n \geq 1$. Recall from Corollary 9.2.16 that the condition

$$n \bar{F}(u_n(x)) \rightarrow -\log H(x), \quad u_n(x) = c_n x + d_n, \quad (9.3.38)$$

for any x in the support of H is sufficient for the convergence result

$$c_n^{-1}(X_{(N_T(n))} - d_n) \xrightarrow{d} Y,$$

where Y has distribution H^λ . An application of condition (9.3.38) to formula (9.3.37) yields a more sophisticated limit relation for any $x_k < \dots < x_1$ in the support of the extreme value distribution H :

$$\begin{aligned} & P(c_n^{-1}(X_{(N_T(n))-i+1}) - d_n) \leq x_i, i = 1, \dots, k) \\ &= e^{-\lambda n \bar{F}(u_n(x_k))} \times \\ & \quad \times \sum_{\ell_2=0}^1 \sum_{\ell_3=0}^{2-\ell_2} \sum_{\ell_4=0}^{3-\ell_3-\ell_2} \dots \sum_{\ell_k=0}^{k-1-\ell_{k-1}-\dots-\ell_2} \prod_{l=0}^k \frac{(\lambda n F(u_n(x_l), u_n(x_{l-1})))^{\ell_l}}{\ell_l!} \cdot \\ & \rightarrow H^\lambda(x_k) \times \\ & \quad \times \sum_{\ell_2=0}^1 \sum_{\ell_3=0}^{2-\ell_2} \sum_{\ell_4=0}^{3-\ell_3-\ell_2} \dots \sum_{\ell_k=0}^{k-1-\ell_{k-1}-\dots-\ell_2} \prod_{l=0}^k \frac{[\log(H^\lambda(x_{l-1})/H^\lambda(x_l))]^{\ell_l}}{\ell_l!}. \end{aligned} \quad (9.3.39)$$

Every extreme value distribution H has a Lebesgue density h . Therefore partial differentiation with respect to all x_i yields the density of the limiting distribution in (9.3.39). The interested reader may derive the form of the density.

We omit further details and simply read off the result from Theorems 4.2.8 and 4.3.4 in Embrechts et al. [46]:

Proposition 9.3.2 (Joint limit distribution of k upper order statistics)

Assume that (9.3.38) holds for any x in the support of an extreme value distribution H and suitable constants $c_n > 0$, $d_n \in \mathbb{R}$, $n \geq 1$. Then for every $k \geq 1$,

$$(c_n^{-1}(X_{(N_T(n)-i+1)} - d_n))_{i=1,\dots,k} \xrightarrow{d} (Y_\lambda^{(i)})_{i=1,\dots,k}, \quad (9.3.40)$$

where the limit vector is the k -dimensional extremal variate corresponding to the extreme value distribution H^λ . It has density

$$h_\lambda(x_1, \dots, x_k) = H^\lambda(x_k) \prod_{i=1}^k \frac{h_\lambda(x_i)}{H^\lambda(x_i)}, \quad x_k < \dots < x_1,$$

where the arguments x_i are chosen from the support of H , $h_\lambda = \lambda h H^{\lambda-1}$ is the Lebesgue density of H^λ and h is the density of H .

For $\lambda = 1$, the extremal variate has density for $x_k < \dots < x_1$ in the support of H given by

$$\begin{aligned} H = \Phi_\alpha: \quad \varphi_\alpha(x_1, \dots, x_k) &= \alpha^k \exp \left\{ -x_k^{-\alpha} - (\alpha + 1) \sum_{j=1}^k \log x_j \right\}, \\ H = \Psi_\alpha: \quad \psi_\alpha(x_1, \dots, x_k) &= \alpha^k \exp \left\{ -(-x_k)^\alpha + (\alpha - 1) \sum_{j=1}^k \log(-x_j) \right\}, \\ H = \Lambda: \quad \lambda(x_1, \dots, x_k) &= \exp \left\{ -e^{-x_k} - \sum_{j=1}^k x_j \right\}. \end{aligned}$$

Notice that the cases $\lambda \neq 1$ and $\lambda = 1$ are closely related. Indeed, since $h_\lambda = \lambda h H^{\lambda-1}$ we have

$$\begin{aligned} h_\lambda(x_1, \dots, x_k) &= \lambda^k H^\lambda(x_k) \prod_{i=1}^k \frac{h(x_i)}{H(x_i)}, \\ &= \lambda^k H^{\lambda-1}(x_k) h_1(x_1, \dots, x_k), \quad x_k < \dots < x_1. \end{aligned} \quad (9.3.41)$$

Example 9.3.3 (The limit distribution of the value of an ECOMOR reinsurance treaty with claim size distribution in the MDA of the Gumbel distribution)

Recall the value of an ECOMOR reinsurance treaty from (9.3.36): for any fixed $k \geq 2$,

$$\begin{aligned}
 R_{\text{ECOMOR}}(n) &= \sum_{i=1}^{k-1} X_{(N_T(n)-i+1)} - (k-1)X_{(N_T(n)-k+1)} \\
 &= \sum_{i=1}^{k-1} (X_{(N_T(n)-i+1)} - X_{(N_T(n)-k+1)}) \\
 &= \sum_{i=1}^{k-1} \sum_{j=i}^{k-1} (X_{(N_T(n)-j+1)} - X_{(N_T(n)-j)}) \\
 &= \sum_{j=1}^{k-1} j (X_{(N_T(n)-j+1)} - X_{(N_T(n)-j)}).
 \end{aligned}$$

Now assume that $F \in \text{MDA}(\Lambda)$ for suitable normalizing and centering constants $c_n > 0$ and $d_n \in \mathbb{R}$. An application of (9.3.40) jointly with the continuous mapping theorem (see Billingsley [17]) yields

$$\begin{aligned}
 c_n^{-1} R_{\text{ECOMOR}}(n) &= \sum_{i=1}^{k-1} i [(c_n^{-1}(X_{(N_T(n)-i+1)} - d_n) - c_n^{-1}(X_{(N_T(n)-i)} - d_n))] \\
 &\xrightarrow{d} \sum_{i=1}^{k-1} i [Y_\lambda^{(i)} - Y_\lambda^{(i+1)}].
 \end{aligned} \tag{9.3.42}$$

Here $(Y_\lambda^{(i)})_{i=1, \dots, k}$ is the extremal variate corresponding to the extreme value distribution Λ^λ . We have

$$(Y_\lambda^{(1)}, \dots, Y_\lambda^{(k)}) \stackrel{d}{=} (Y_1^{(1)} + \log \lambda, \dots, Y_1^{(k)} + \log \lambda). \tag{9.3.43}$$

This equality can be checked by comparison of the densities on both sides, using relation (9.3.41). We conclude that the limit distribution in (9.3.42) does not depend on λ . Therefore we assume without loss of generality that $\lambda = 1$.

One way to get more information about the limit distribution in (9.3.42) is to choose some special distribution $F \in \text{MDA}(\Lambda)$ for which one can calculate the distribution of $c_n^{-1} R_{\text{ECOMOR}}(n)$ explicitly for every $n \geq 1$. Such an F is the standard exponential distribution. Direct calculation shows that

$$n \bar{F}(x + \log n) \rightarrow e^{-x}, \quad x \in \mathbb{R}.$$

Hence we can choose $c_n = 1$ and $d_n = \log n$. Relation (9.3.42) then yields for $\lambda = 1$

$$R_{\text{ECOMOR}}(n) \xrightarrow{d} \sum_{i=1}^{k-1} i [Y_1^{(i)} - Y_1^{(i+1)}].$$

On the other hand, we know from Example 3.4.3 that for an iid standard exponential claim size sequence $R_{\text{ECOMOR}}(n)$ has a $\Gamma(k-1, 1)$ distribution. This is the limit distribution in (9.3.42).

The asymptotic theory for an ECOMOR treaty with claim size distribution $F \in \text{MDA}(\Delta)$ is rather attractive. The limit distribution is well known. Moreover, the limit relation (9.3.42) does not require knowledge of the Poisson intensity λ and the centering constants d_n . If we know the distribution F we can read off the normalizing constants c_n for some of the standard distributions in the MDA of the Gumbel distribution; see for example Table 9.2.14. $\square$

Example 9.3.4 (The limit distribution of the largest claims reinsurance treaty with claim size distribution in the MDA of the Gumbel distribution) It follows from the arguments given in Example 9.3.3 that for $F \in \text{MDA}(\Delta)$ one can find constants $c_n > 0$ and $d_n \in \mathbb{R}$ such that

$$c_n^{-1}(R_{\text{LC}}(n) - k d_n) = \sum_{i=1}^k [c_n^{-1}(X_{(N_T(n)-i+1)} - d_n)] \\ \xrightarrow{d} \sum_{i=1}^k Y_\lambda^{(i)} \stackrel{d}{=} \sum_{i=1}^k Y_1^{(i)} + k \log \lambda, \quad (9.3.44)$$

where $(Y_\lambda^{(1)}, \dots, Y_\lambda^{(k)})$ is the extremal variate corresponding to the extreme value distribution Λ^λ . We assume without loss of generality that $\lambda = 1$.

In order to identify the limit distribution in (9.3.44) we consider the special case of an iid standard exponential claim size sequence. We know from Example 9.3.3 that we can choose $c_n = 1$ and $d_n = \log n$. We recall from (3.4.49) that

$$R_{\text{LC}}(n) - k \log n \xrightarrow{d} \sum_{i=1}^k E_i + k \sum_{i=k+1}^{\infty} i^{-1}(E_i - 1) + k C_k = R, \quad (9.3.45)$$

where (E_i) is an iid standard exponential sequence, $C_k = \gamma - \sum_{i=1}^k i^{-1}$, and $\gamma = 0.5772 \dots$ is Euler's constant.

The limit distribution R in (9.3.45) is rather unfamiliar. Writing

$$R_1 = \sum_{i=1}^k E_i \quad \text{and} \quad R_2 = k \sum_{i=k+1}^{\infty} i^{-1}(E_i - 1),$$

we see that the distribution of R is determined by the constant $k C_k$ and the independent summands R_1 and R_2 . Of course, R_1 has a $\Gamma(k, 1)$ distribution. The characteristic function of R can be easily evaluated and from it one can derive the moments. An elementary calculation yields

$$ER = k + k C_k \sim -k \log k, \quad k \rightarrow \infty, \\ \text{var}(R) = k + k \sum_{i=k+1}^{\infty} i^{-2} \sim k, \quad k \rightarrow \infty.$$

Thus the expectation and variance of R for large k are essentially determined by $kC_k \sim -k \log k$ and $\text{var}(R_1) = k$, respectively. For more detailed quantitative analyses of the distribution of R it is useful to employ simulations. Figure 3.4.5 gives one an idea about the density of the distribution of R . $\square$

Comments

Extreme value theory has natural applications in non-life insurance, in particular in reinsurance. This aspect is stressed in Embrechts et al. [46]. In the presence of large claims, extreme value theory advises one *not* to use the classical approach to collective risk theory by applying mean value and variance analyses of a portfolio or asymptotic analyses based on the strong law of large numbers and the central limit theorem. The study of the asymptotic behavior of the reinsurance treaties of extreme value type above shows that rather unfamiliar distributional limits may occur. In Embrechts et al. [46], Section 8.7, an asymptotic theory is also developed for these treaties when the underlying claim size distribution is in the maximum domain of attraction of the Fréchet distribution. However, the resulting limits are even more complicated than those in $\text{MDA}(\Lambda)$.

Exercises

- (1) Let (E_i) be an iid standard exponential sequence. Show that the exponential *spacings* $E_{(n-i+1)} - E_{(n-i)}$ of the ordered sample $E_{(1)} < \dots < E_{(n)}$ a.s. are mutually independent for any $n \geq 1$ and that the random variables $i(E_{(n-i+1)} - E_{(n-i)})$ are iid with common standard exponential distribution.

Hint: Start by solving Exercise 13 on p. 49 which gives a representation of the ordered sample. Then apply a linear transformation to this vector and the density transformation theorem.

- (2) Prove relation (9.3.43) by using (9.3.41) for $H = \Lambda$.